

Cape Romano-Ten Thousand Islands Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

Contents

Indicators	2
Nutrients	2
Total Nitrogen - Discrete	2
Total Phosphorus - Discrete	4
Water Quality	6
Dissolved Oxygen - Continuous	6
Dissolved Oxygen - Discrete	8
Dissolved Oxygen Saturation - Continuous	10
Dissolved Oxygen Saturation - Discrete	12
Salinity - Discrete	14
Salinity - Continuous	15
Water Temperature - Continuous	19
Water Temperature - Discrete	24
pH - Continuous	26
pH - Discrete	28
Specific Conductivity - Continuous	30
Water Clarity	32
Turbidity - Continuous	32
Turbidity - Discrete	34
Total Suspended Solids - Discrete	36
Chlorophyll a, Uncorrected for Pheophytin - Discrete	38
Chlorophyll a, Corrected for Pheophytin - Discrete	40
Secchi Depth - Discrete	42

Indicators

Nutrients

Total Nitrogen - Discrete

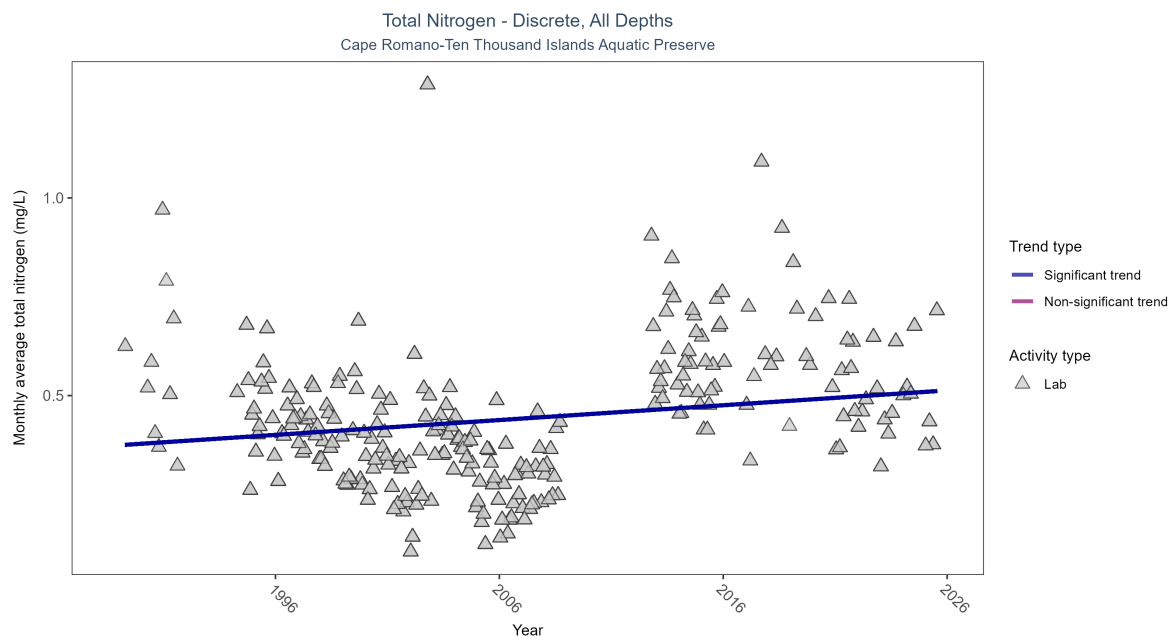


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	2023	32	1989 - 2025	0.38645	0.13153	0.37438	0.00375	0.0024

Monthly average total nitrogen increased by less than 0.01 mg/L per year.

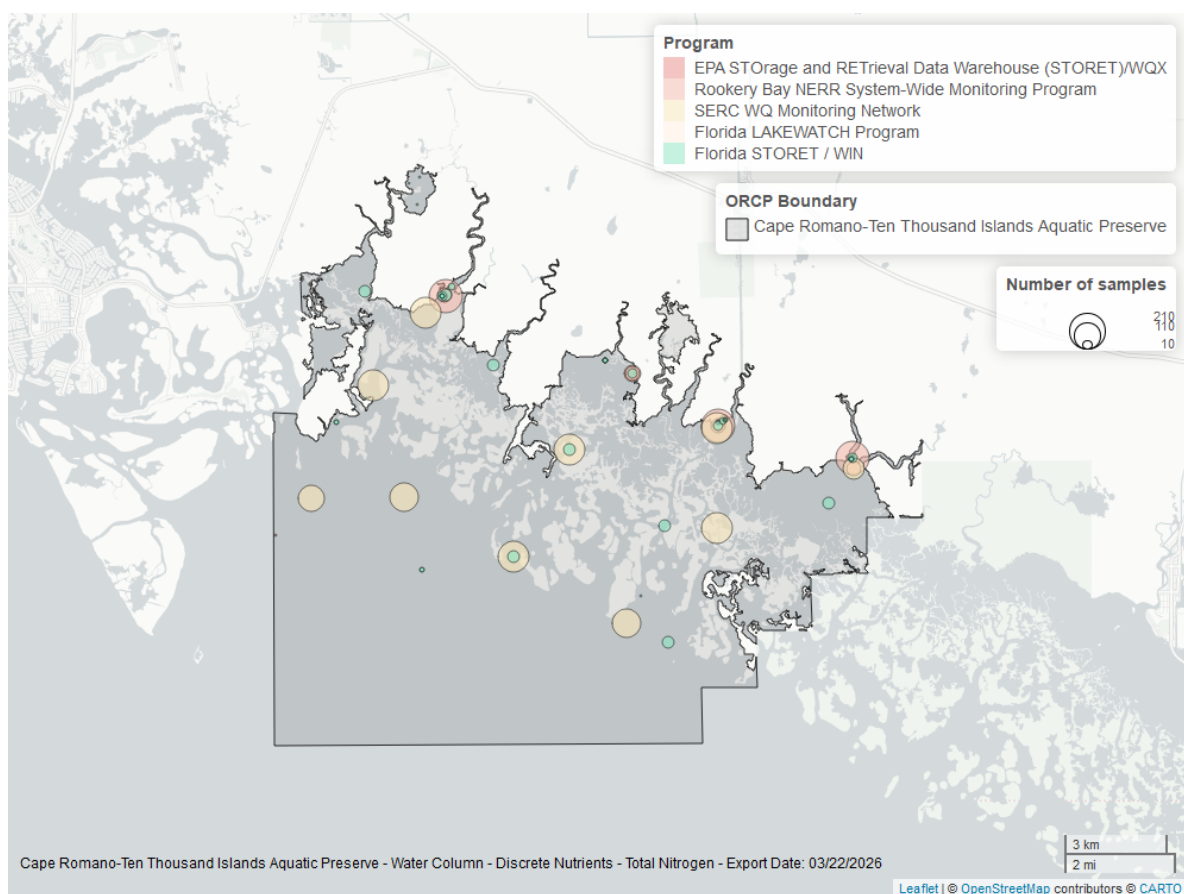


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

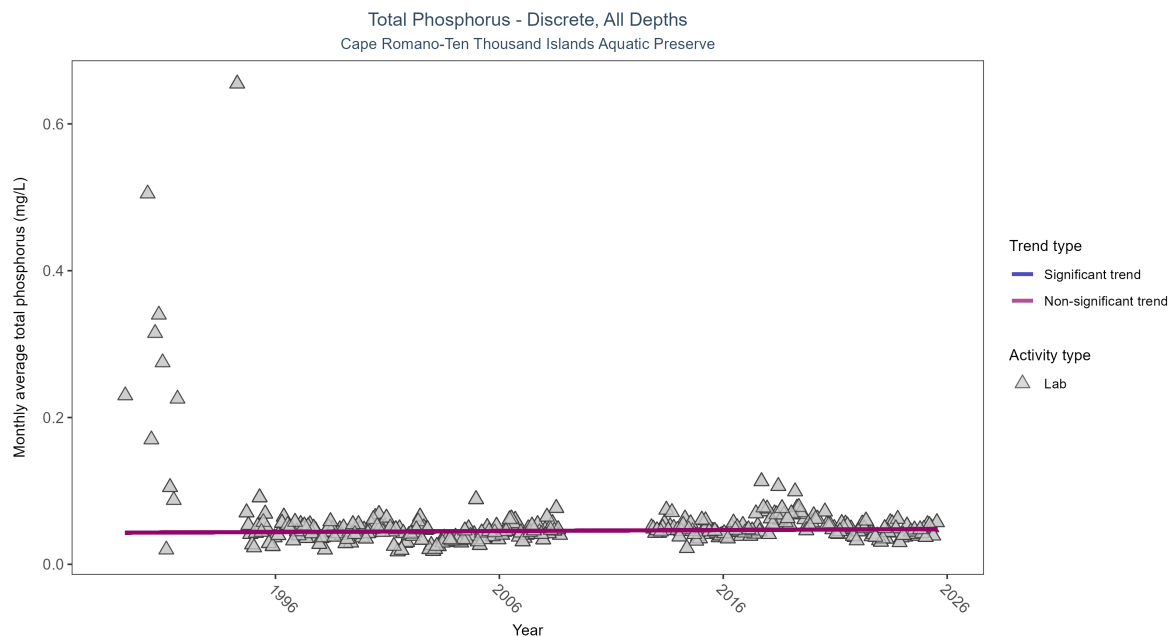


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	2407	32	1989 - 2025	0.04345	0.08052	0.04309	0.00014	0.0505

Total phosphorus showed no detectable trend between 1989 and 2025.

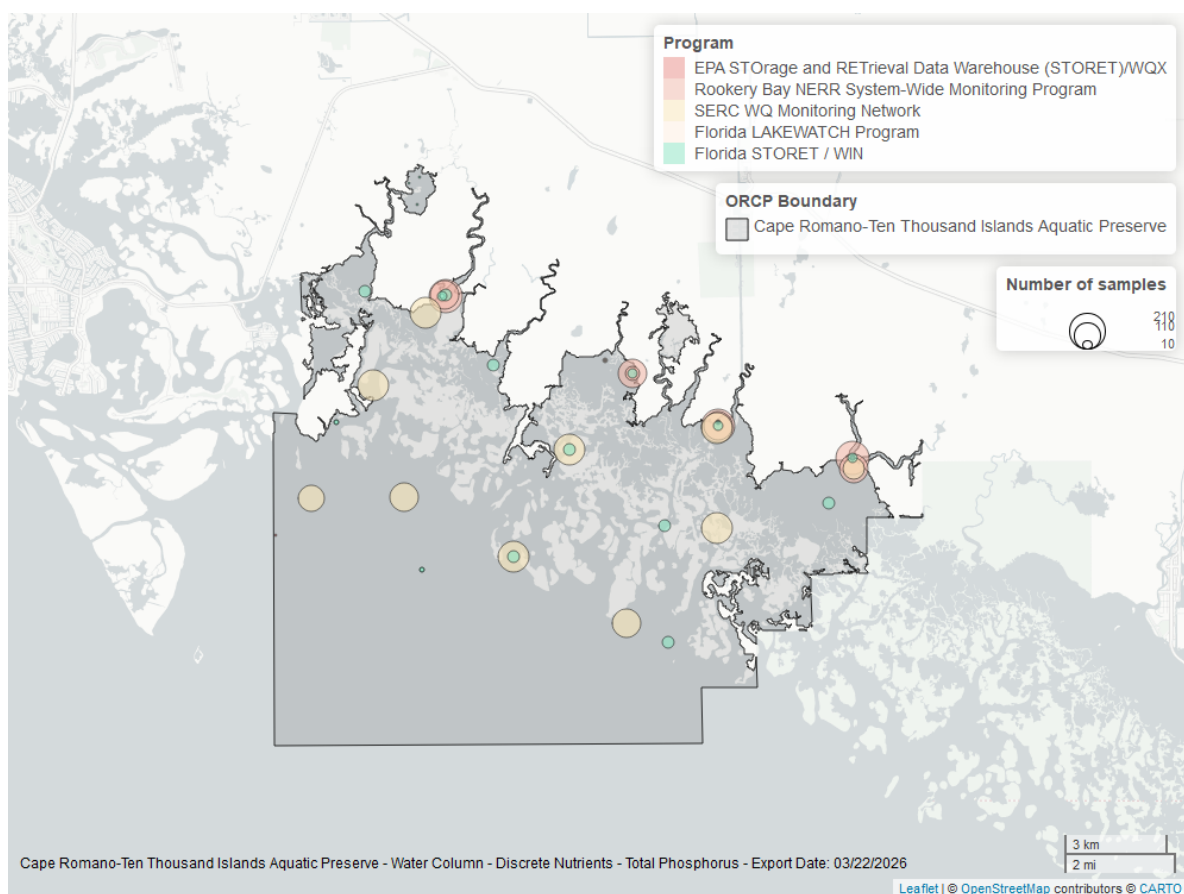


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

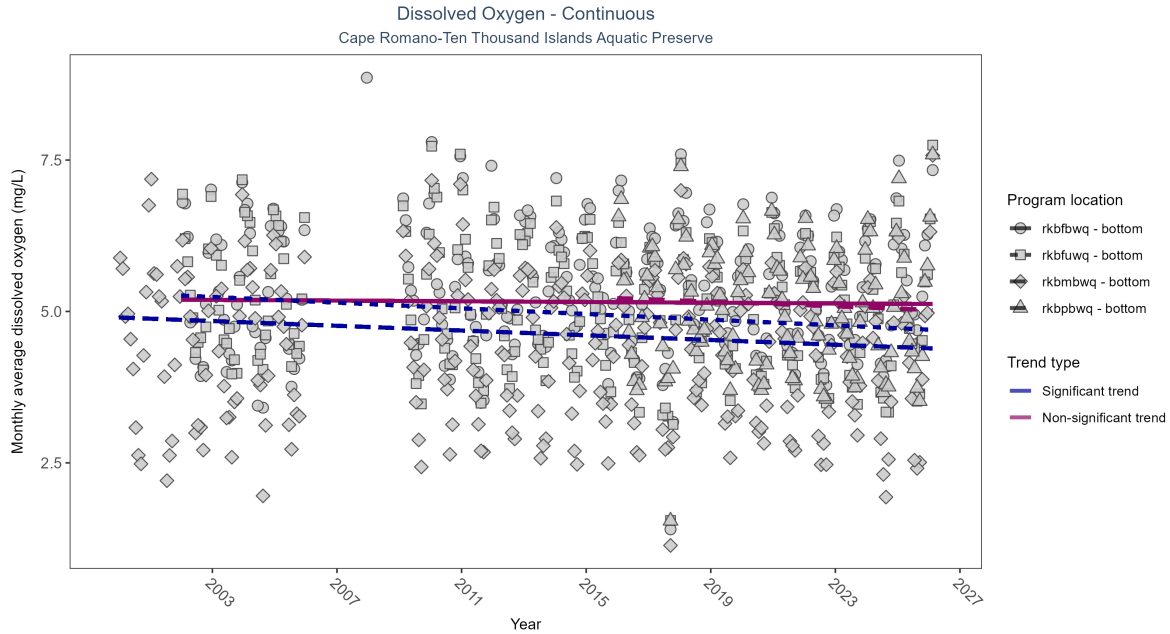


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwbq	No significant trend	603909	23	2002 - 2026	5.4	-0.02	5.19	0.00	0.6197
rkbfuwq	Significantly decreasing trend	636515	22	2002 - 2026	5.1	-0.29	5.26	-0.02	0
rkmbwbq	Significantly decreasing trend	657119	24	2000 - 2026	4.4	-0.24	4.90	-0.02	0
rkbpbwq	No significant trend	332812	11	2016 - 2026	4.9	-0.14	5.22	-0.02	0.0762

At two program locations, monthly average dissolved oxygen decreased by 0.02 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at two locations.

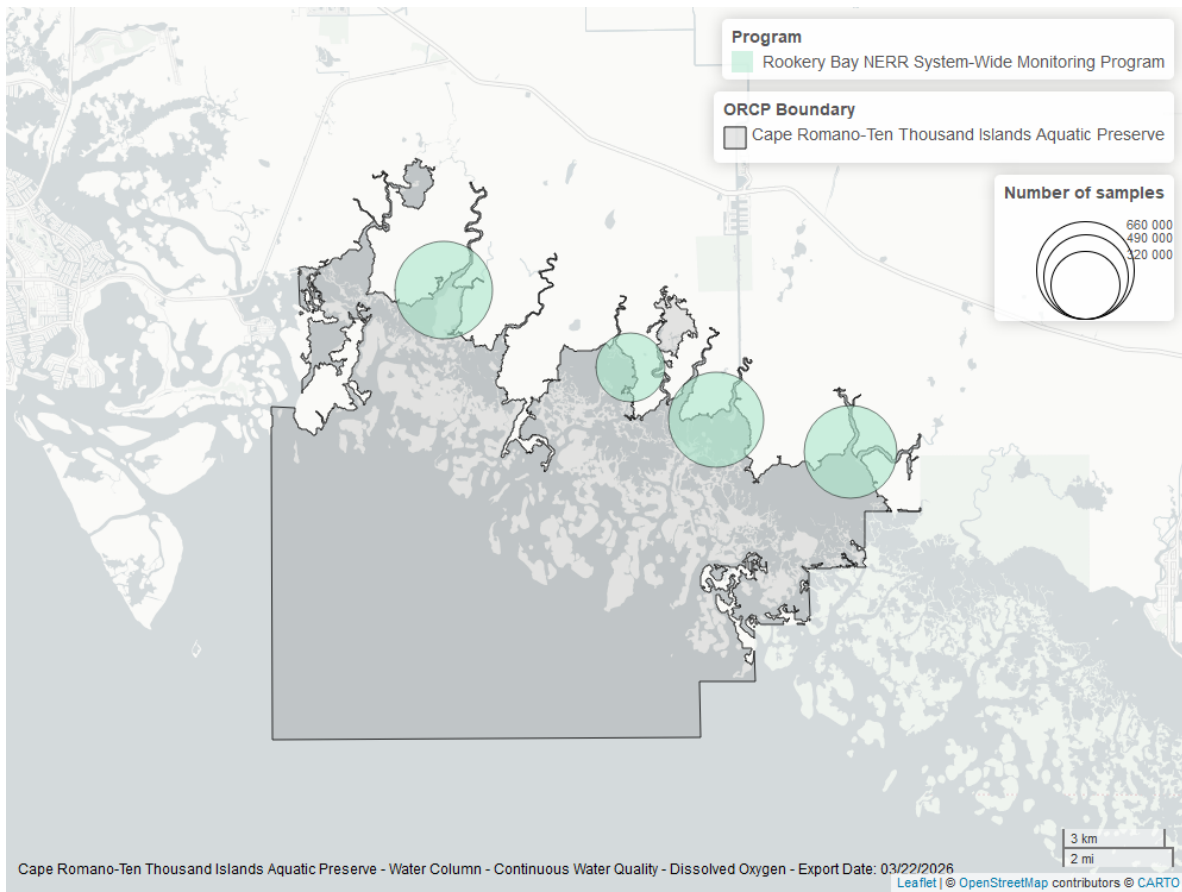


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

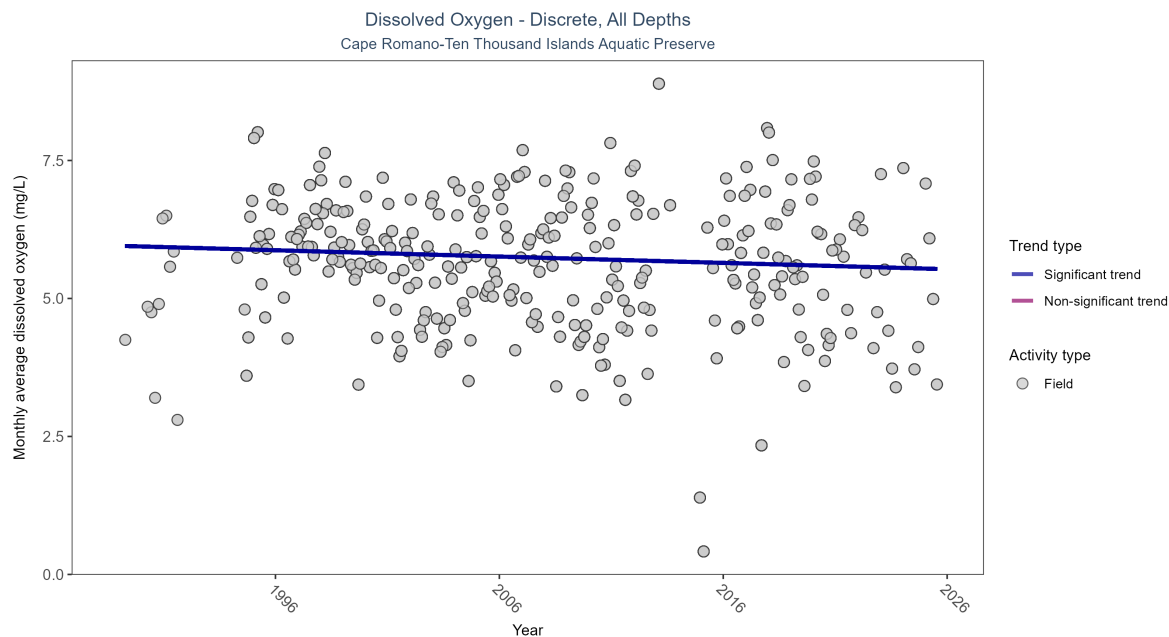


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	10119	35	1989 - 2025	5.8	-0.07697	5.95376	-0.01145	0.0498

Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

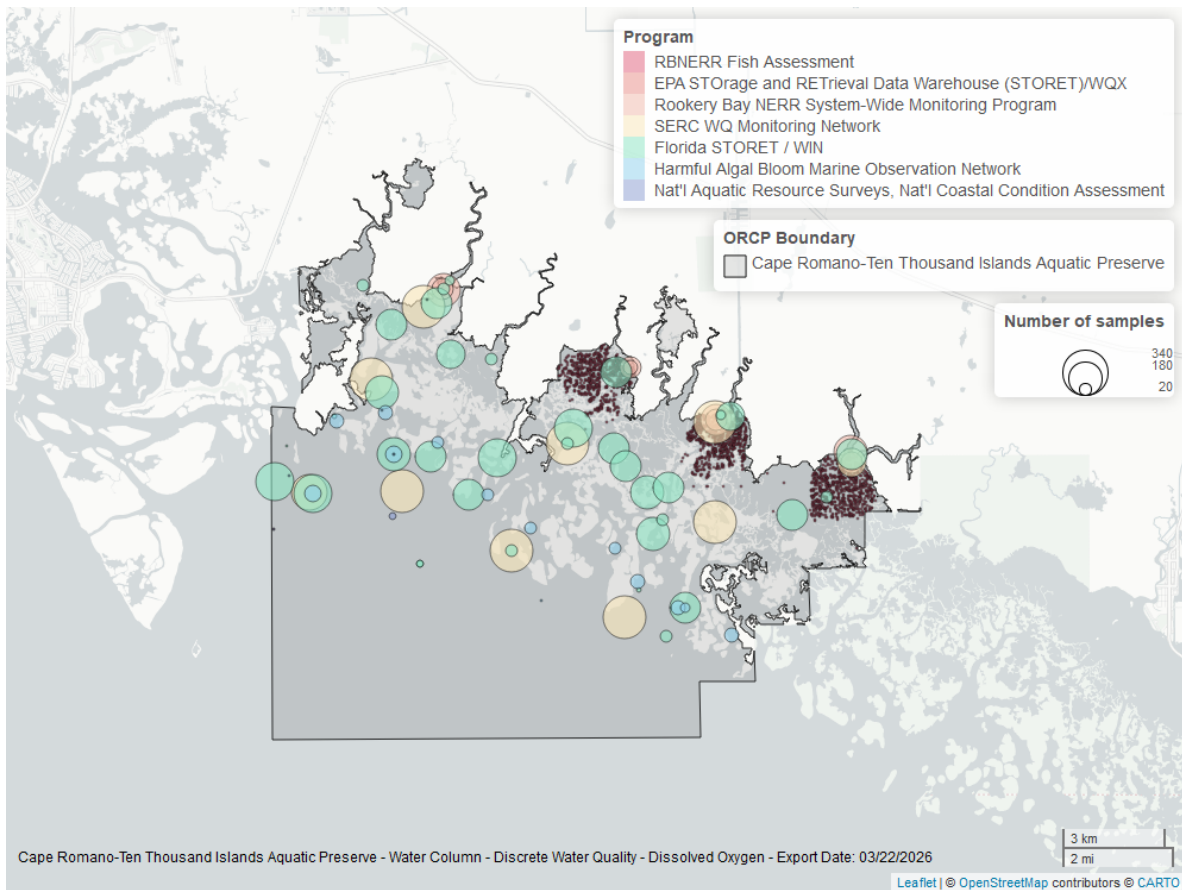


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

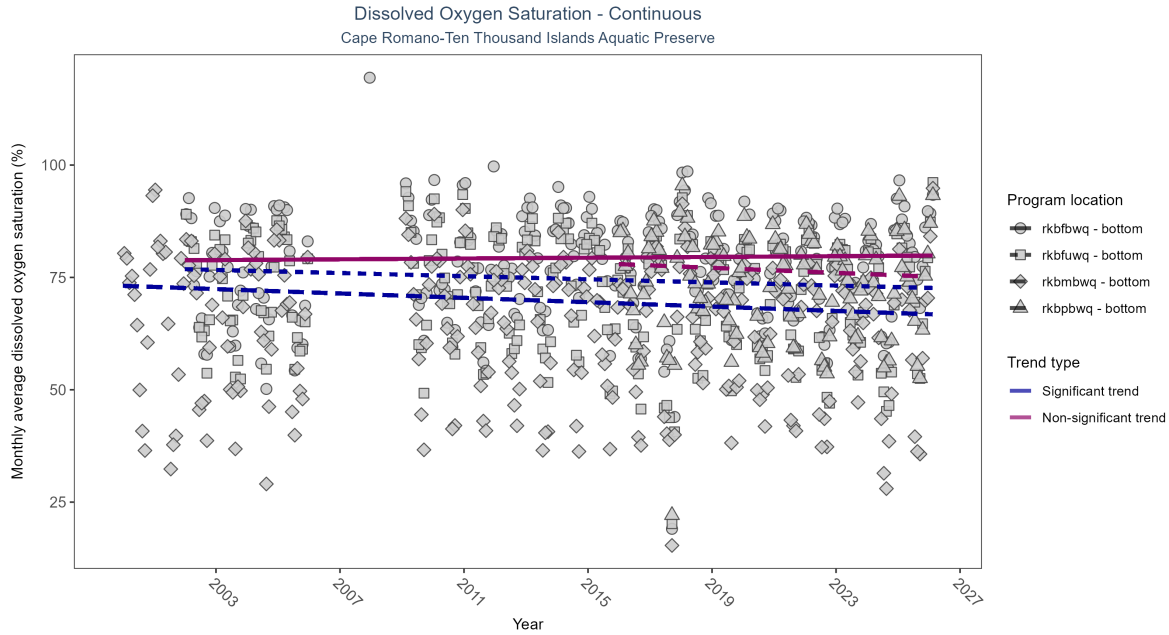


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	No significant trend	608215	23	2002 - 2026	79.0	0.04	78.81	0.04	0.4281
rkbfuwq	Significantly decreasing trend	636774	22	2002 - 2026	72.6	-0.20	76.84	-0.17	0
rkmbwq	Significantly decreasing trend	663443	24	2000 - 2026	65.3	-0.21	73.13	-0.24	0
rkbpwq	No significant trend	336249	11	2016 - 2026	72.7	-0.13	77.96	-0.28	0.0762

At two program locations, monthly average dissolved oxygen saturation decreased by 0.17% per year at one site and by 0.24% per year at the other. No detectable change in monthly average dissolved oxygen saturation was observed at two locations.

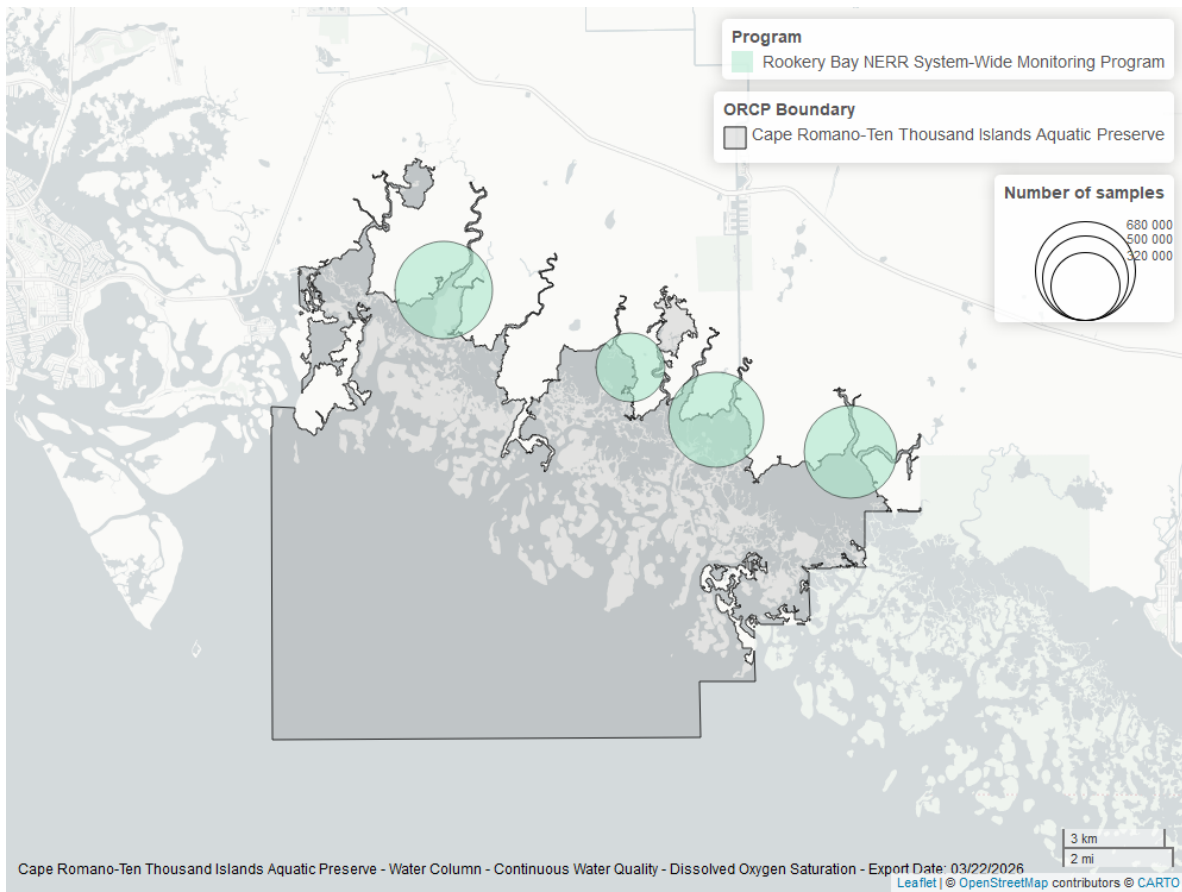


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

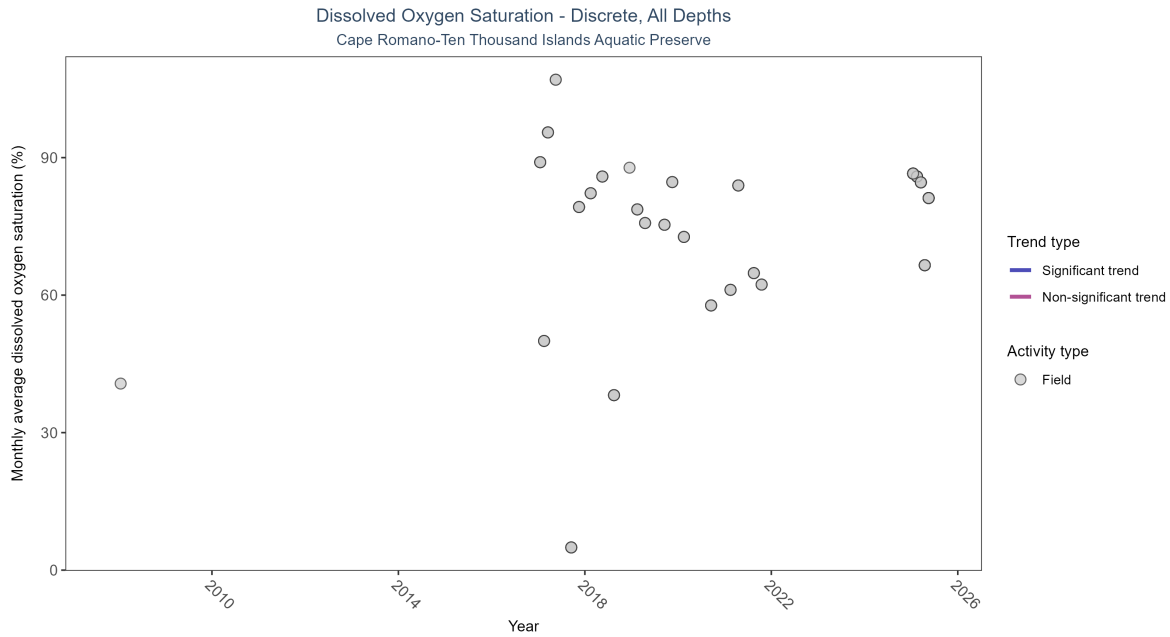


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data to calculate trend	112	7	2008 - 2025	78.1	-	-	-	-

There was insufficient data to fit a model for dissolved oxygen saturation.

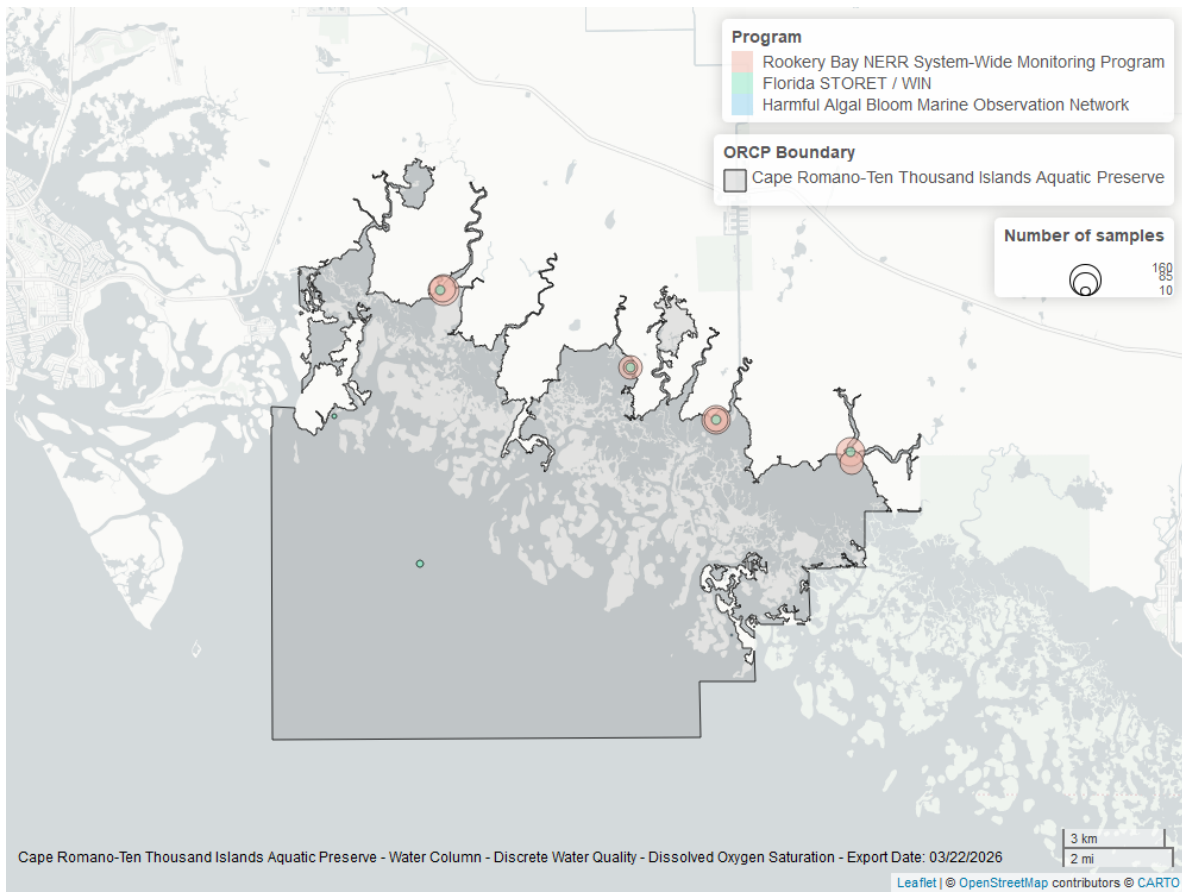


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

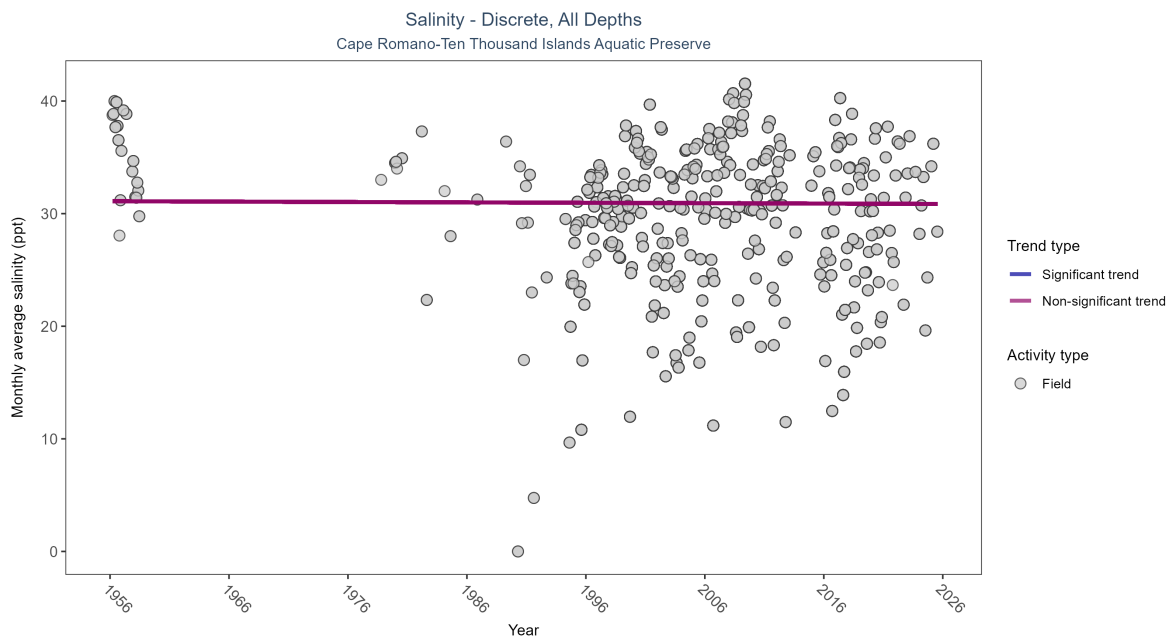


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No significant trend	11080	45	1956 - 2025	31.9	-0.00448	31.10154	-0.00346	0.931

Salinity showed no detectable trend between 1956 and 2025.

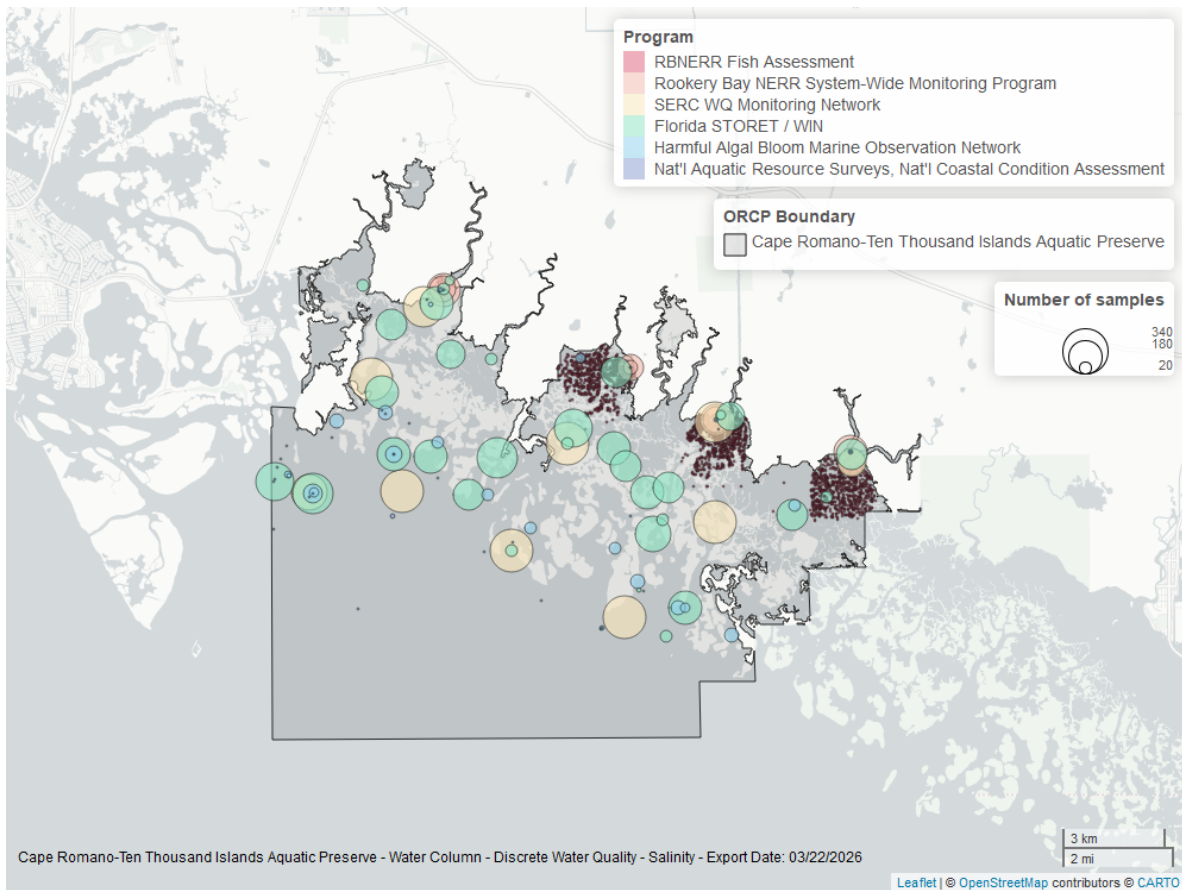


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program -
354

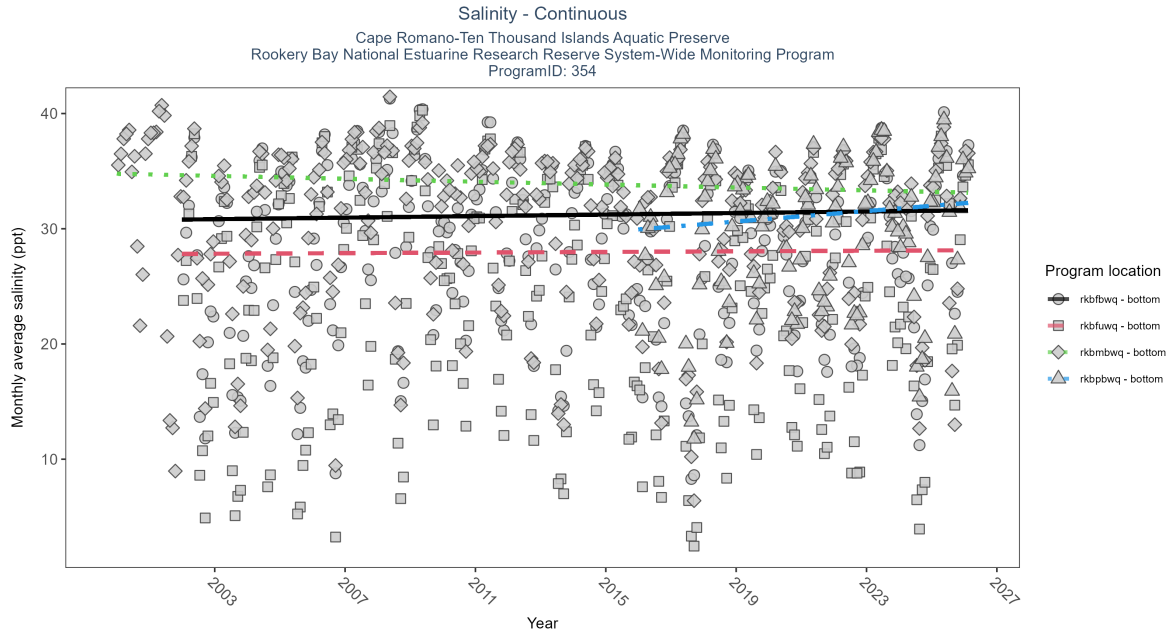


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results for All Stations - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfbwq	No significant trend	703182	25	2002 - 2026	29.9	0.04	30.80	0.03	0.3375
rkbfuwq	No significant trend	717069	25	2002 - 2026	26.3	0.01	27.82	0.01	0.7856
rkmbwq	Significantly decreasing trend	736129	27	2000 - 2026	33.3	-0.10	34.78	-0.06	0.016
rkbpbwq	No significant trend	333127	11	2016 - 2026	30.5	0.14	29.94	0.23	0.0606

At one program location, monthly average salinity increased by 0.31 ppt per year. At two program locations, monthly average salinity decreased by 0.06 ppt per year at one site and by 0.44 ppt per year at the other. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for two locations.

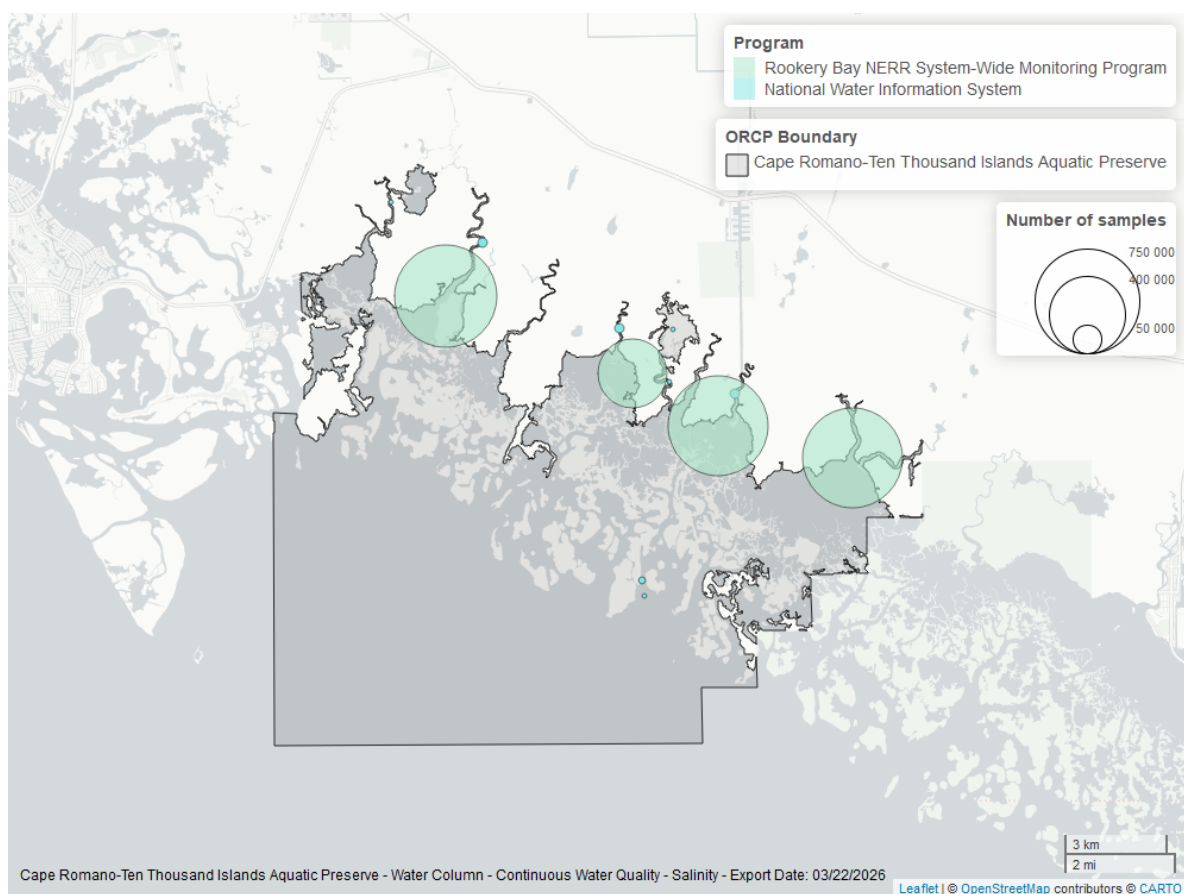


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

National Water Information System - 7

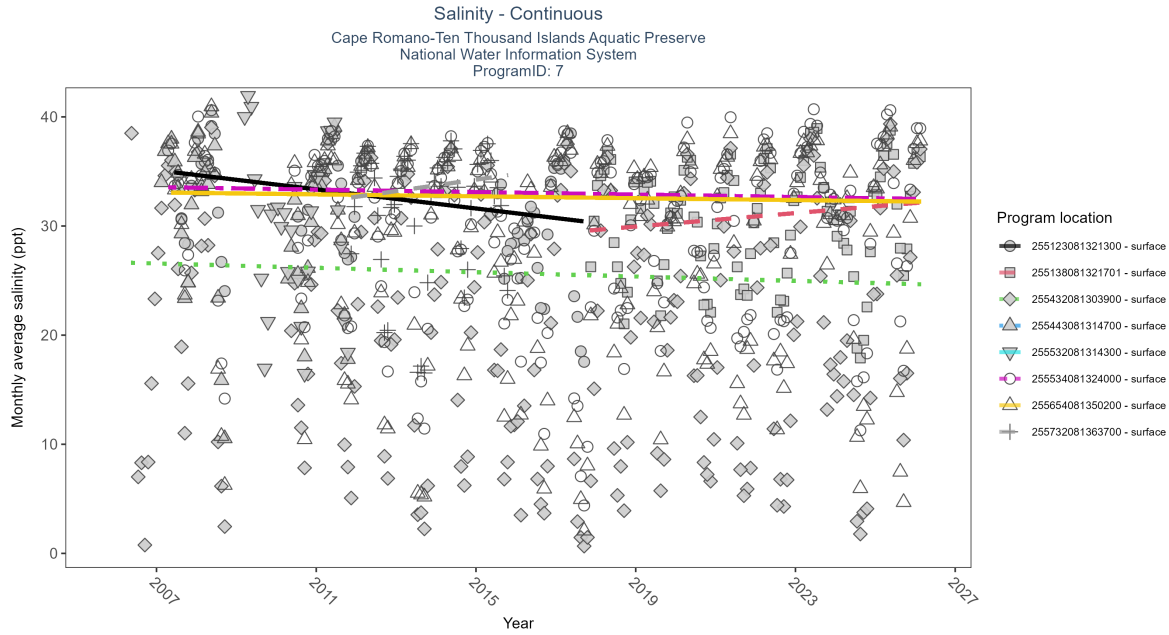


Figure 17: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results for All Stations - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	Significantly decreasing trend	1809	8	2007 - 2017	32	-0.23	35.11	-0.44	0.0182
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	32	0.17	29.33	0.31	0.0294
255432081303900	No significant trend	6468	20	2006 - 2026	22	-0.08	26.67	-0.1	0.1135
255443081314700	Insufficient data to calculate trend	1465	4	2007 - 2011	32	-	-	-	-
255532081314300	Insufficient data to calculate trend	902	3	2009 - 2011	31	-	-	-	-
255534081324000	No significant trend	6162	19	2007 - 2026	32	-0.04	33.56	-0.06	0.3647
255654081350200	No significant trend	6183	19	2007 - 2026	32	-0.04	33.07	-0.04	0.365
255732081363700	No significant trend	1434	5	2011 - 2015	34	0.25	32.15	0.52	0.0955

At one program location, monthly average salinity increased by 0.31 ppt per year. At two program locations, monthly average salinity decreased by 0.06 ppt per year at one site and by 0.44 ppt per year at the other. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for two locations.

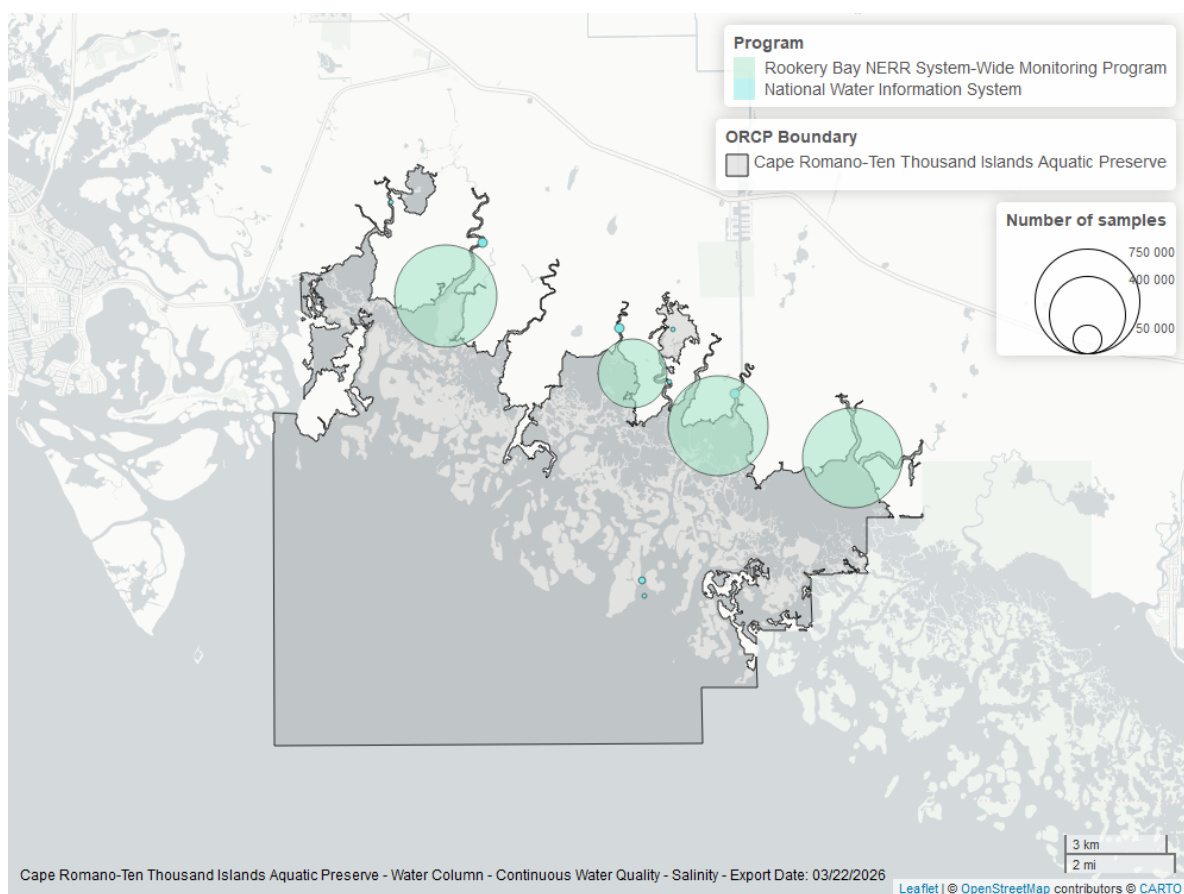


Figure 18: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program -
354

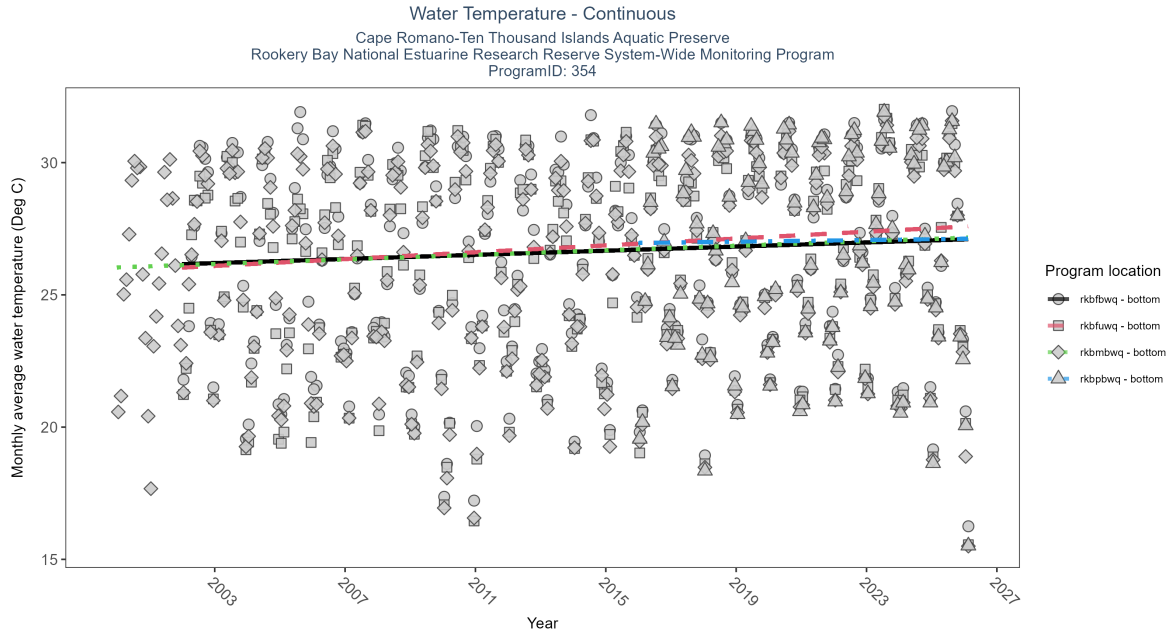


Figure 19: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: Seasonal Kendall-Tau Results for All Stations - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfbwq	Significantly increasing trend	719021	25	2002 - 2026	27.0	0.22	26.16	0.04	0
rkbfuwq	Significantly increasing trend	727501	25	2002 - 2026	26.9	0.26	26.03	0.06	0
rkbbmwq	Significantly increasing trend	762033	27	2000 - 2026	26.8	0.25	26.04	0.04	0
rkbpbwq	No significant trend	337833	11	2016 - 2026	27.4	0.05	26.96	0.02	0.5544

At six program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at four locations. There was insufficient data to fit a model for two locations.

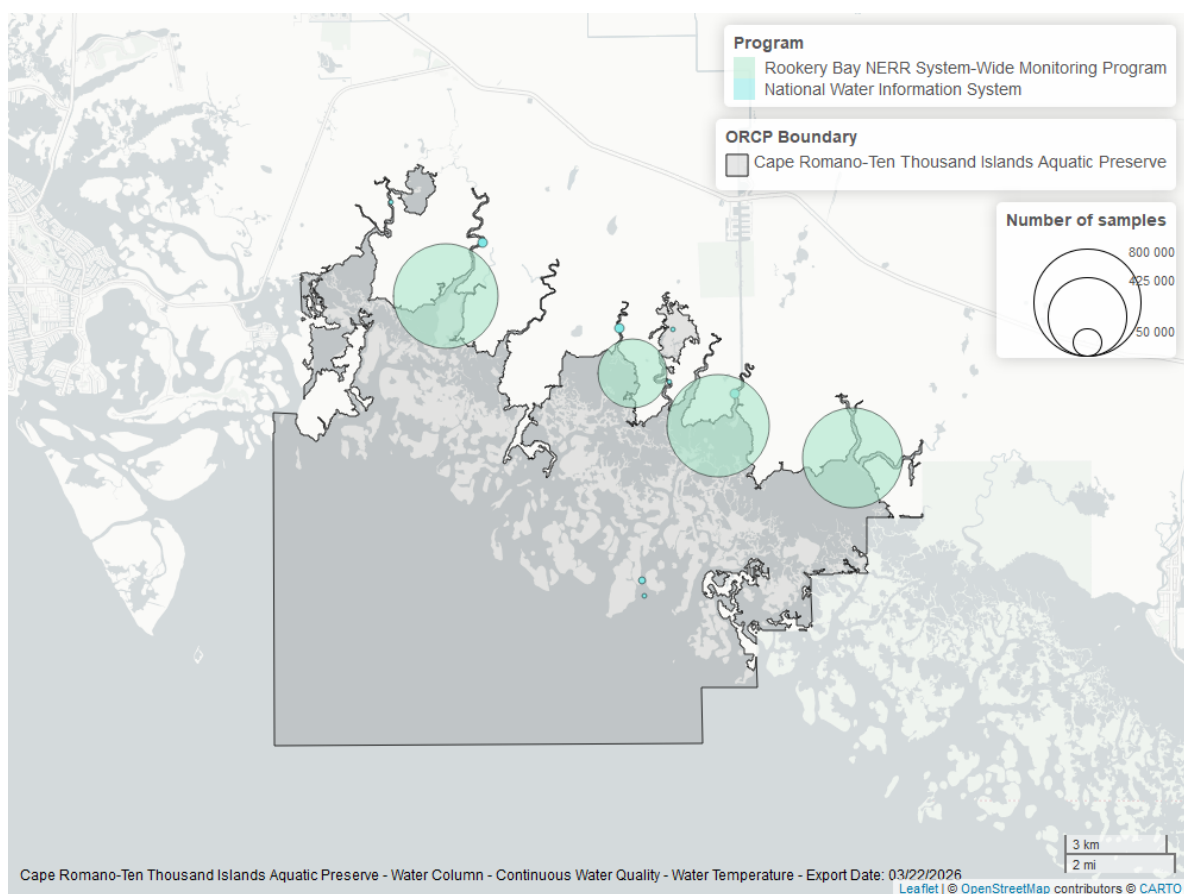


Figure 20: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

National Water Information System - 7

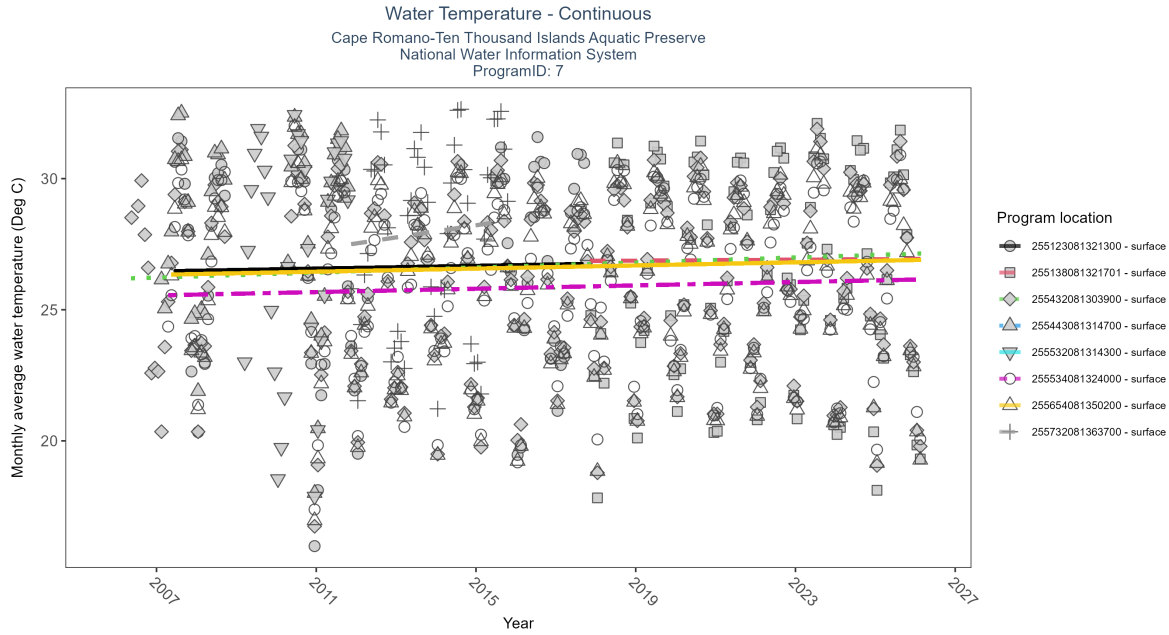


Figure 21: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results for All Stations - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1818	8	2007 - 2017	27.3	0.1	26.46	0.03	0.3843
255138081321701	No significant trend	3005	10	2017 - 2026	27.0	0.02	26.85	0.01	0.9199
255432081303900	Significantly increasing trend	6527	20	2006 - 2026	27.0	0.17	26.18	0.05	0.0005
255443081314700	Insufficient data to calculate trend	2011	4	2007 - 2011	29.3	-	-	-	-
255532081314300	Insufficient data to calculate trend	906	3	2009 - 2011	29.3	-	-	-	-
255534081324000	Significantly increasing trend	6225	19	2007 - 2026	26.6	0.14	25.55	0.03	0.0069
255654081350200	Significantly increasing trend	6222	19	2007 - 2026	26.7	0.11	26.33	0.03	0.0327
255732081363700	No significant trend	1435	5	2011 - 2015	28.4	0.24	27.3	0.23	0.1149

At six program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at four locations. There was insufficient data to fit a model for two locations.

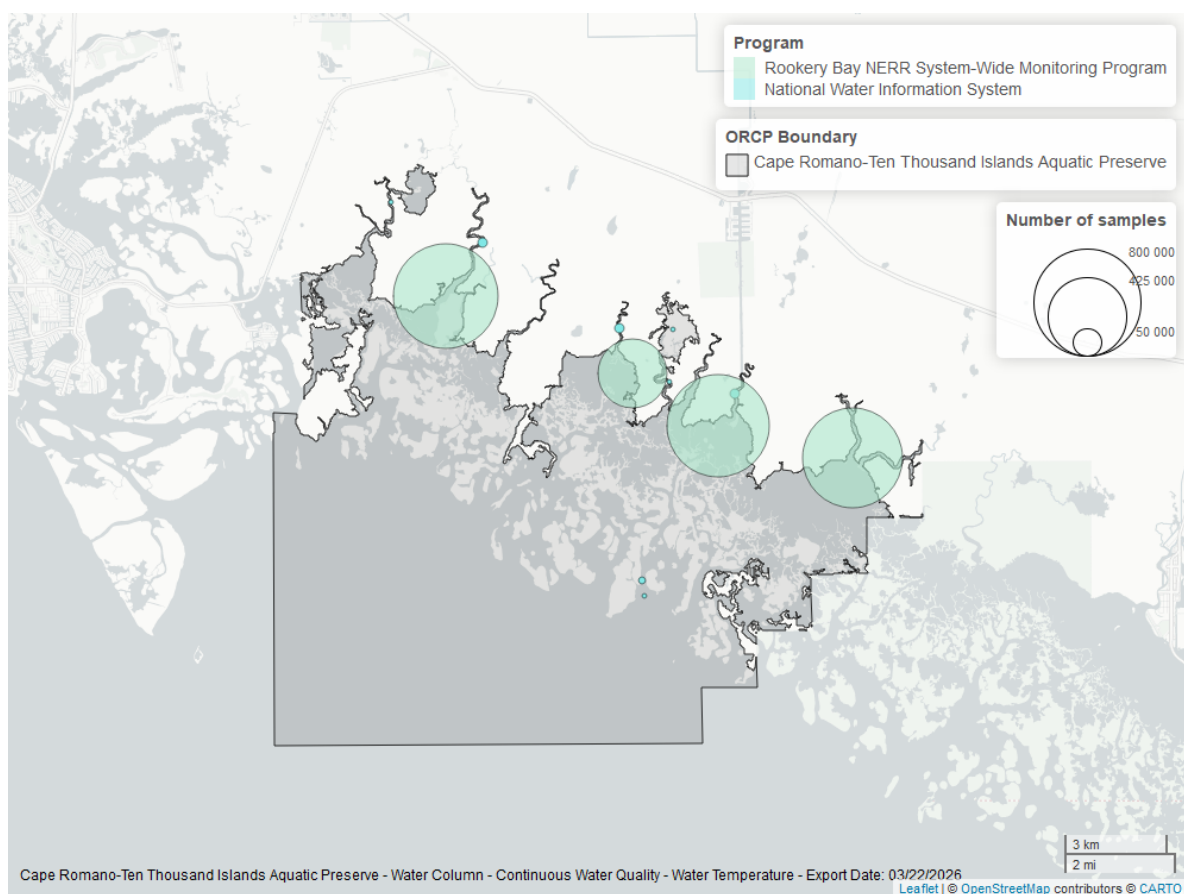


Figure 22: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

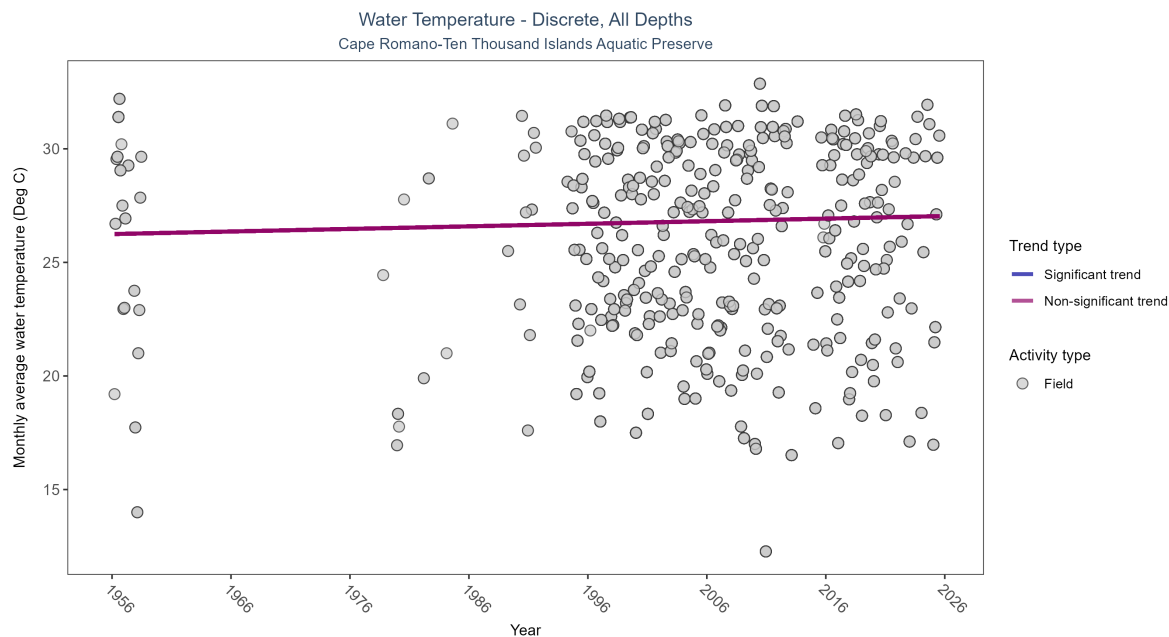


Figure 23: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	10694	43	1956 - 2025	26.8	0.06277	26.24631	0.01134	0.0788

Water temperature showed no detectable trend between 1956 and 2025.

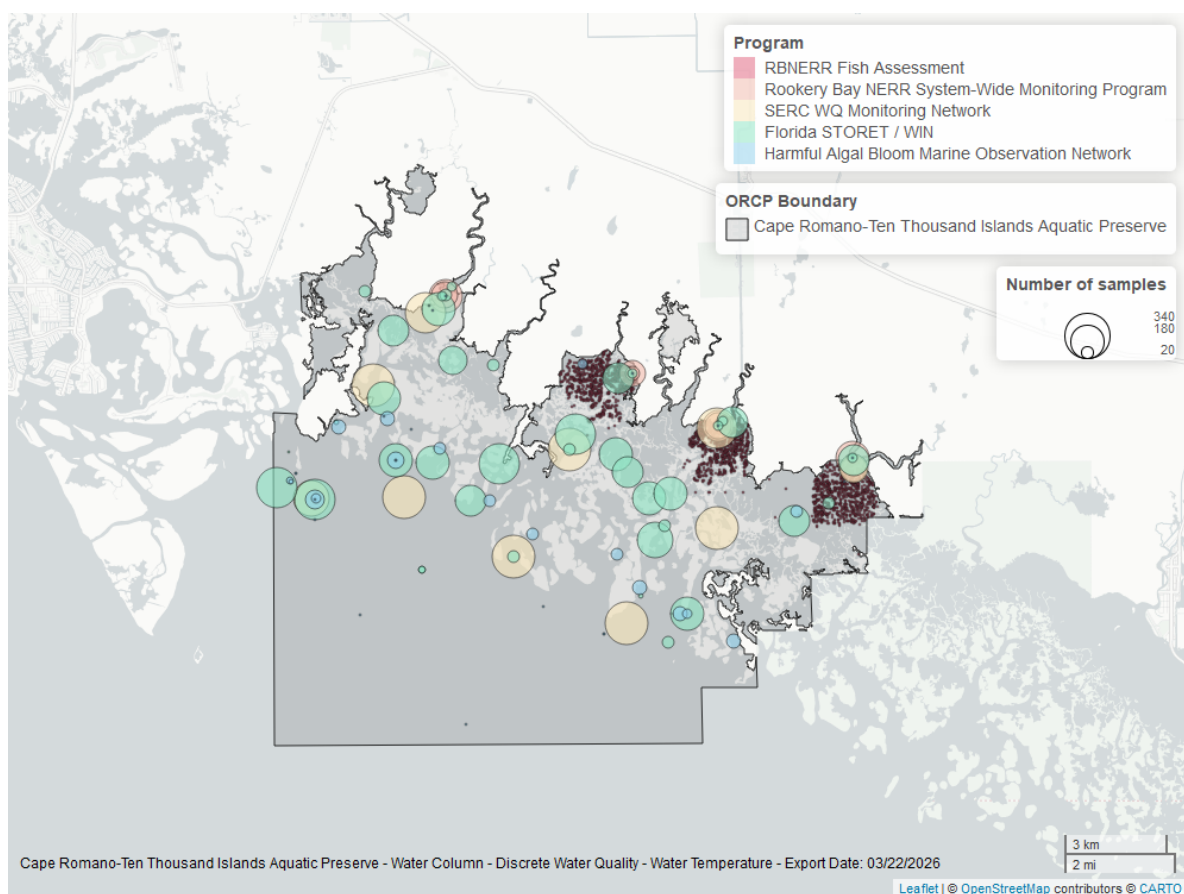


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

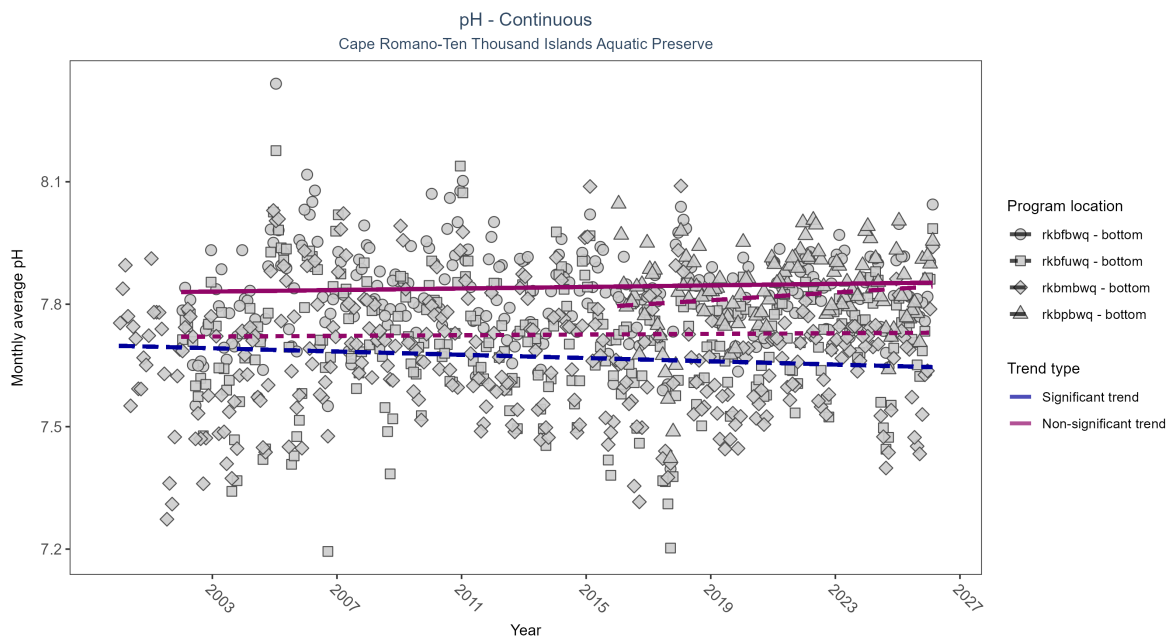


Figure 25: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwbq	No significant trend	670684	25	2002 - 2026	7.8	0.05	7.83	0	0.2078
rkbfuwq	No significant trend	689115	25	2002 - 2026	7.7	0.02	7.72	0	0.6096
rkmbwbq	Significantly decreasing trend	727383	27	2000 - 2026	7.7	-0.10	7.70	0	0.0184
rkbpbwq	No significant trend	328113	11	2016 - 2026	7.8	0.13	7.80	0	0.0762

At one program location, monthly average pH decreased by less than 0.01 pH units per year. No detectable change in monthly average pH was observed at three locations.

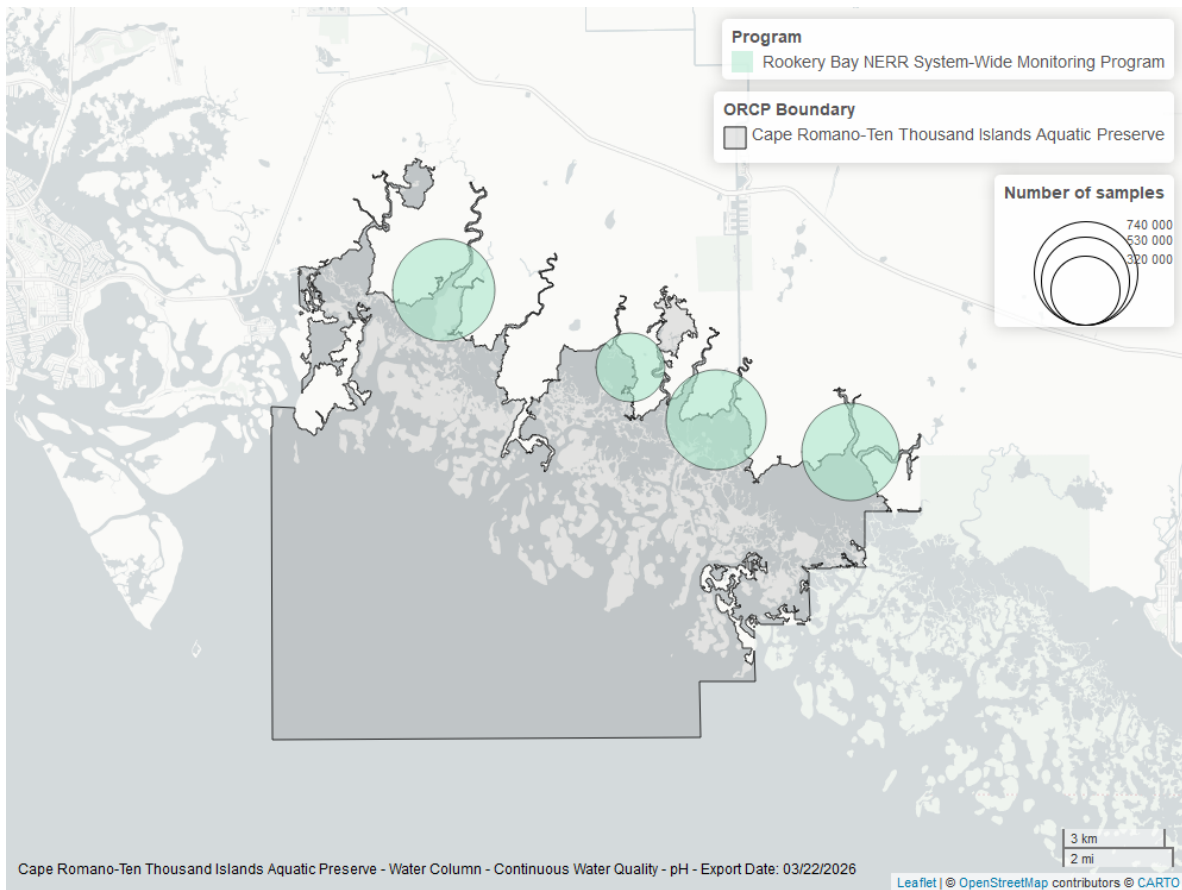


Figure 26: Map showing location of pH continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

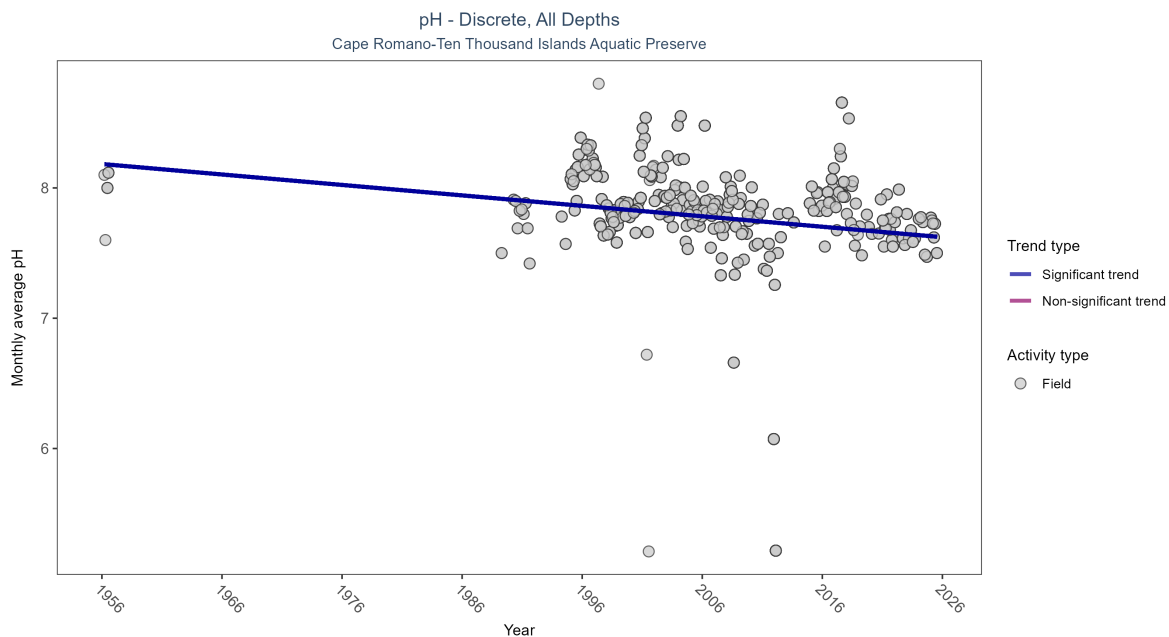


Figure 27: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	4301	36	1956 - 2025	7.9	-0.23696	8.18343	-0.00802	0

Monthly average pH decreased by 0.01 pH units per year.

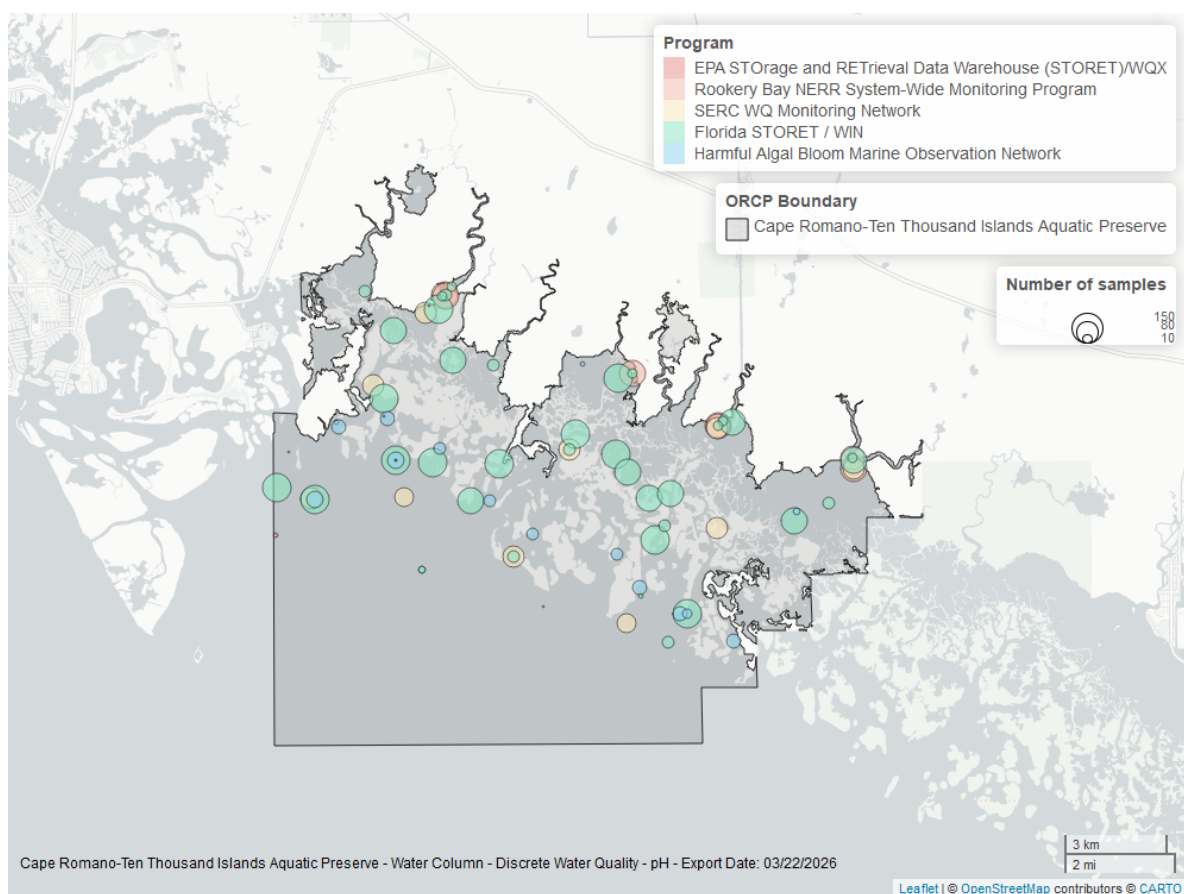


Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

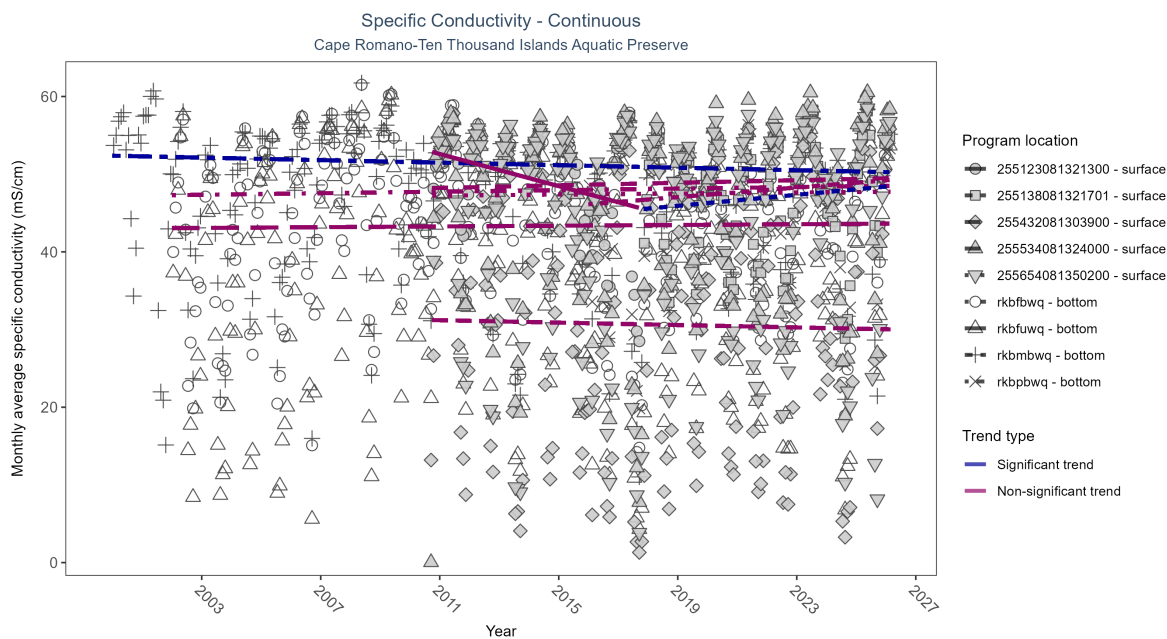


Table 15: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1295	6	2010 - 2017	49.10	-0.20	53.63	-1.03	0.2034
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	48.30	0.18	45.21	0.36	0.0248
255432081303900	No significant trend	5512	17	2010 - 2026	34.40	-0.03	31.28	-0.08	0.6149
255534081324000	No significant trend	5509	17	2010 - 2026	48.60	0.04	48.14	0.08	0.4303
255654081350200	No significant trend	5547	17	2010 - 2026	48.60	0.03	47.21	0.03	0.5649
rkbfwq	No significant trend	703174	25	2002 - 2026	46.07	0.04	47.30	0.05	0.3304
rkbfuwq	No significant trend	717061	25	2002 - 2026	41.17	0.01	43.06	0.02	0.7747
rkmbwq	Significantly decreasing trend	736129	27	2000 - 2026	50.83	-0.09	52.37	-0.08	0.0197
rkbpwq	No significant trend	333114	11	2016 - 2026	47.03	0.13	46.12	0.31	0.068

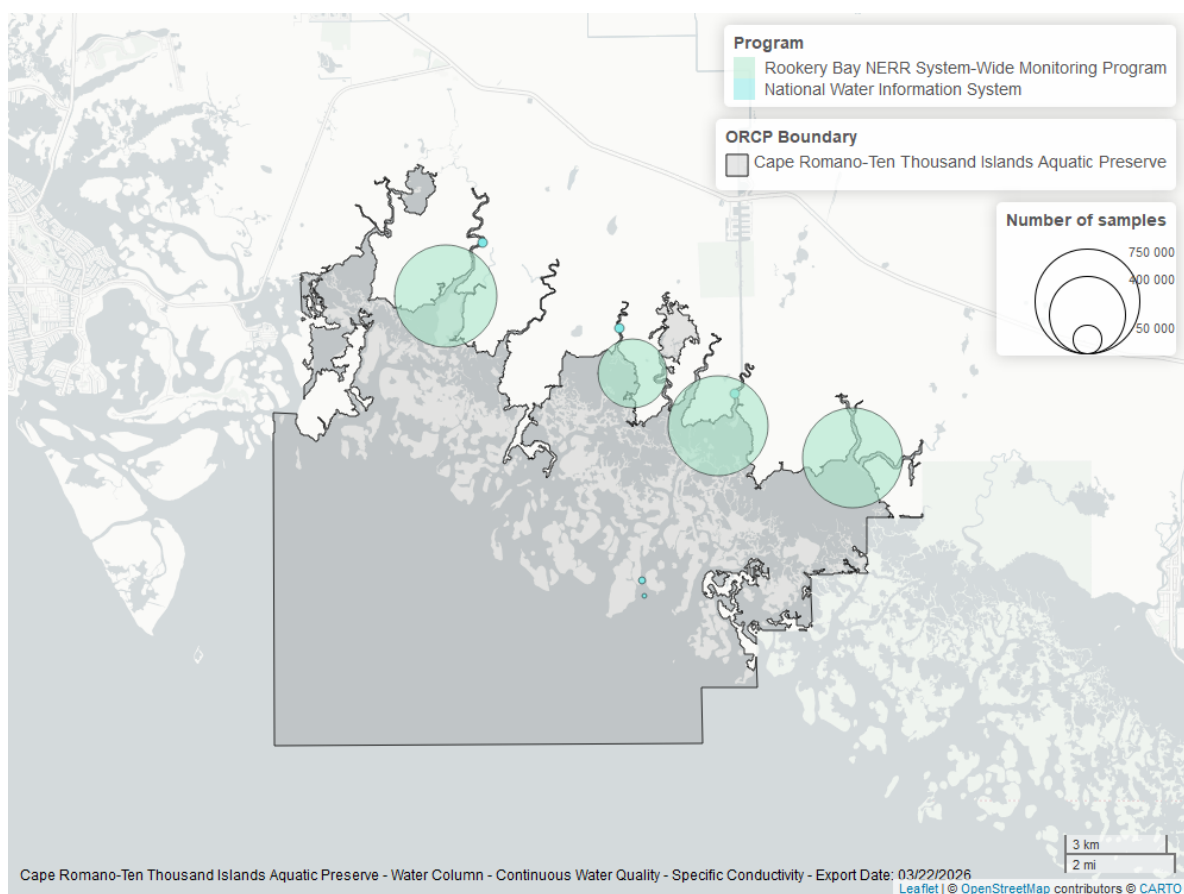


Figure 29: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

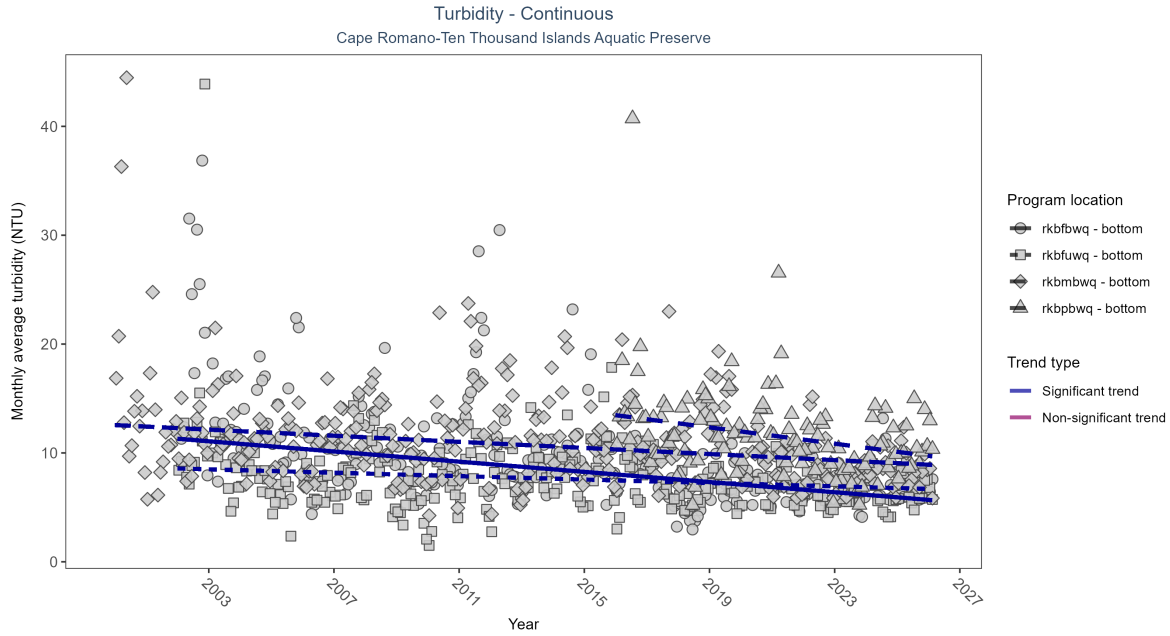


Figure 30: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 16: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwbq	Significantly decreasing trend	682915	25	2002 - 2026	7	-0.40	11.30	-0.23	0
rkbfuwq	Significantly decreasing trend	657778	25	2002 - 2026	6	-0.19	8.57	-0.08	0
rkmbwbq	Significantly decreasing trend	714237	27	2000 - 2026	9	-0.24	12.56	-0.14	0
rkbpbwq	Significantly decreasing trend	331299	11	2016 - 2026	9	-0.31	13.47	-0.37	0

At four program locations, monthly average turbidity decreased between 0.08 and 0.37 NTU per year.

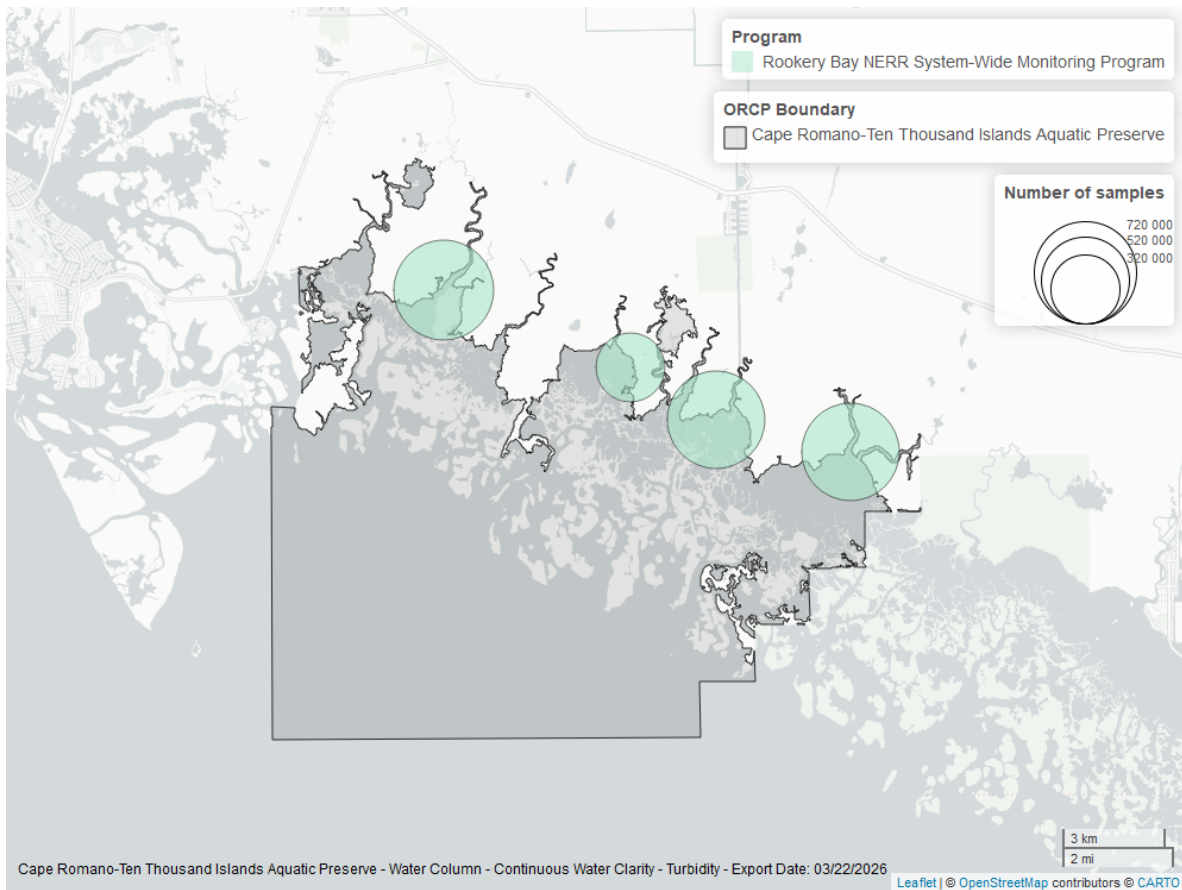


Figure 31: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

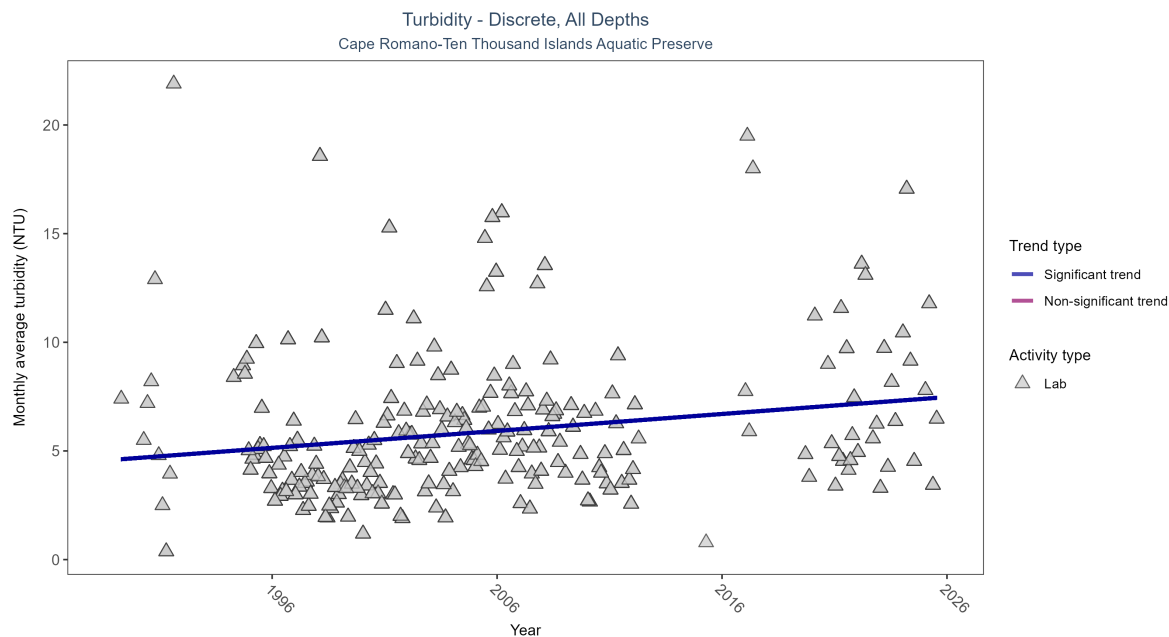


Figure 32: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	4677	31	1989 - 2025	4	0.15654	4.59643	0.07792	0.0005

Monthly average turbidity increased by 0.08 NTU per year, indicating a decrease in water clarity.

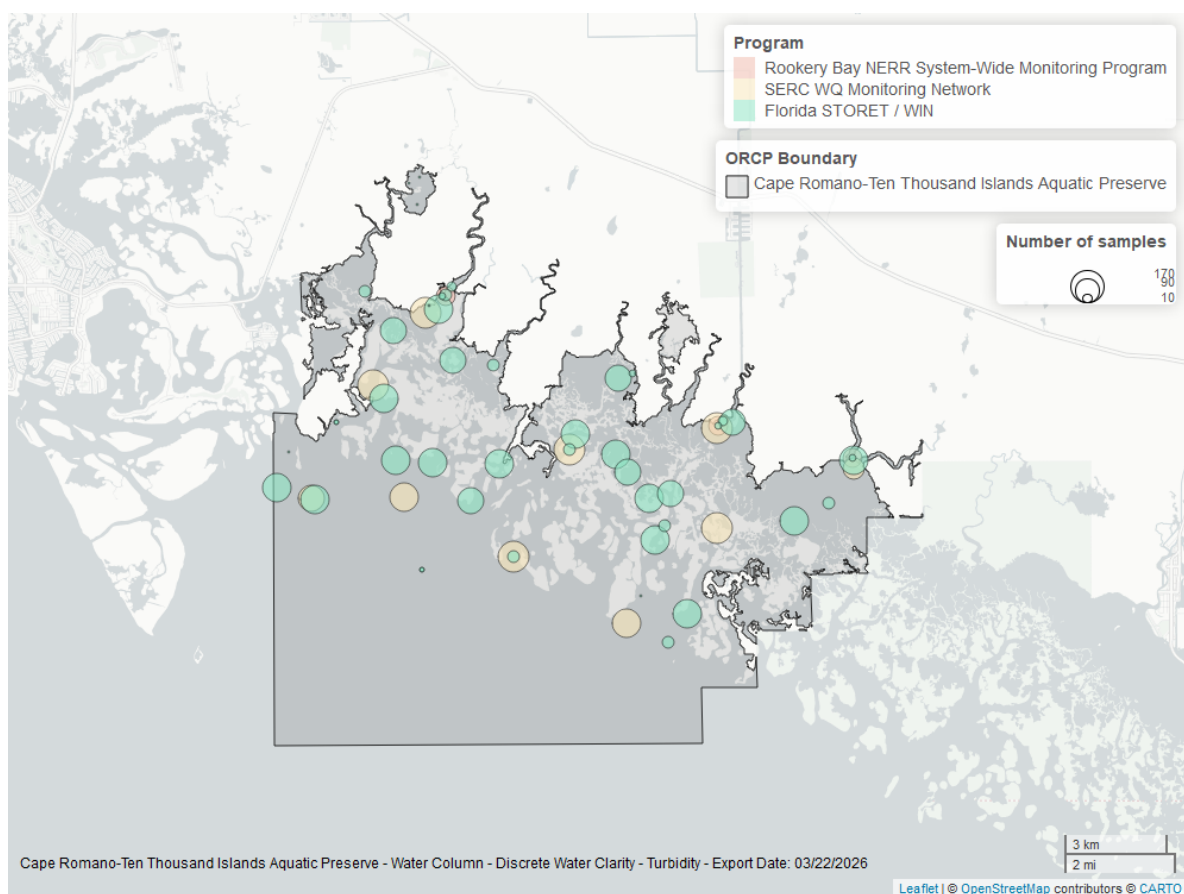


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

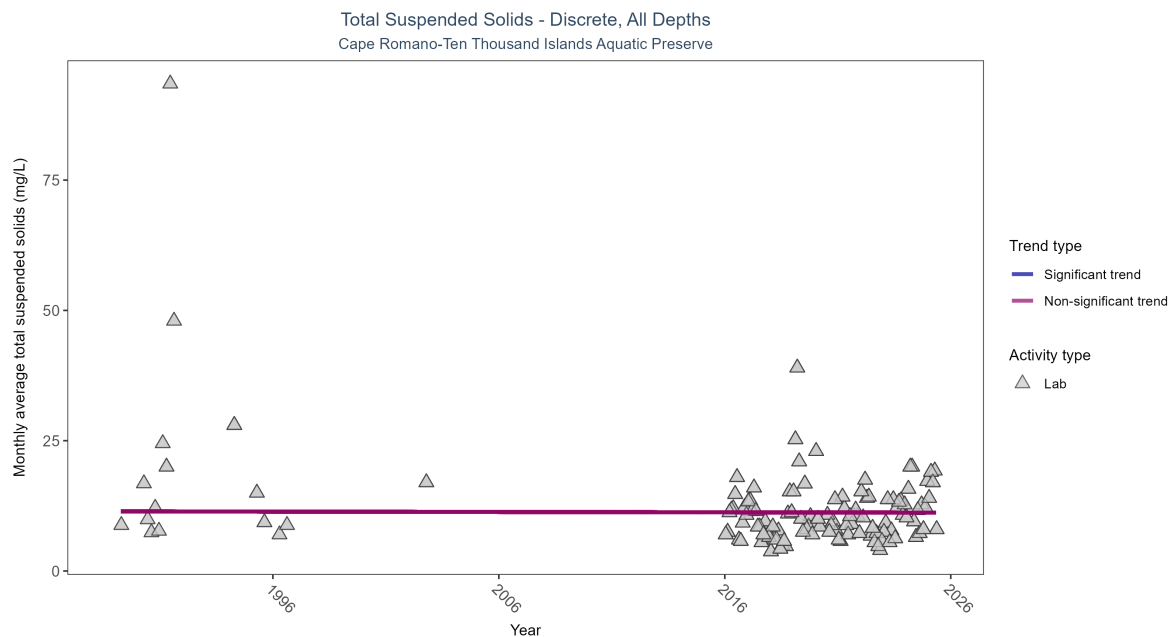


Figure 34: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	484	17	1989 - 2025	10	-0.00639	11.43828	-0.00667	0.8532

Total suspended solids showed no detectable trend between 1989 and 2025.

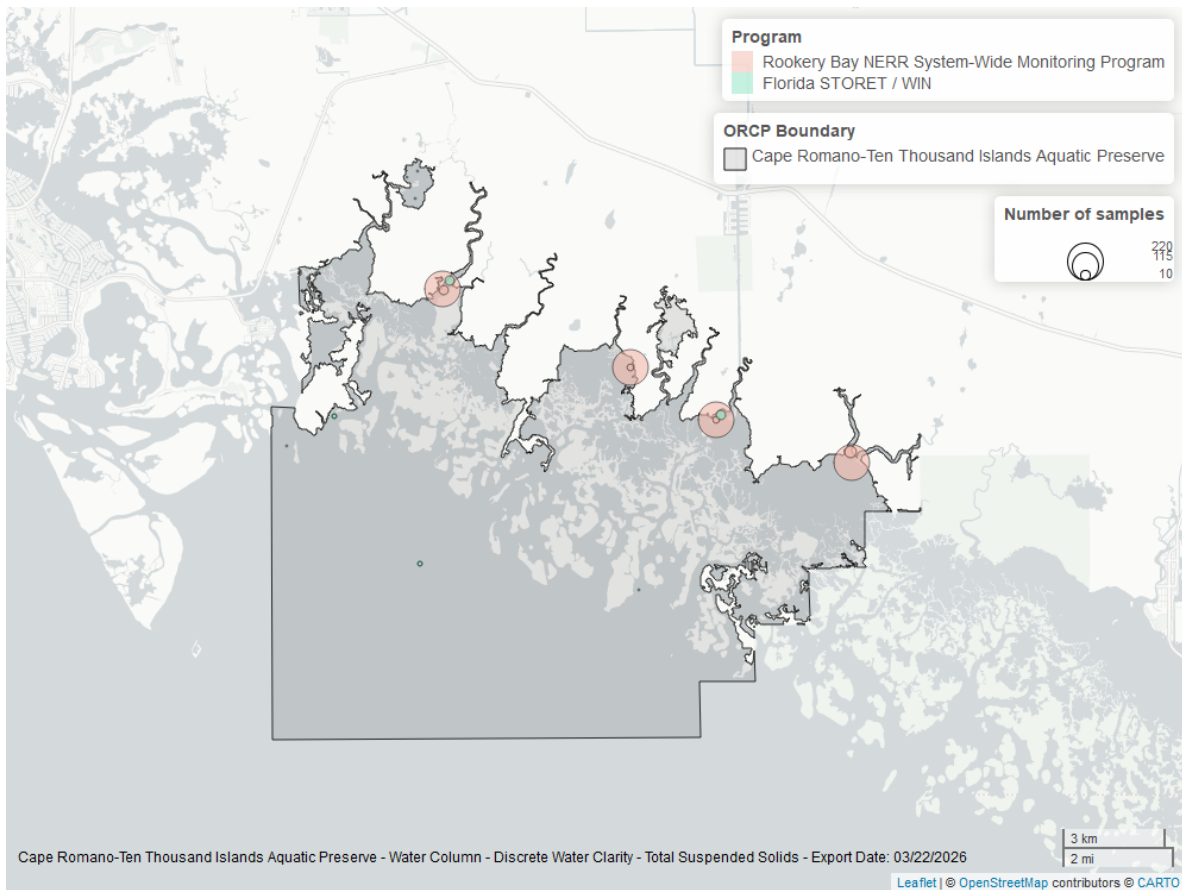


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

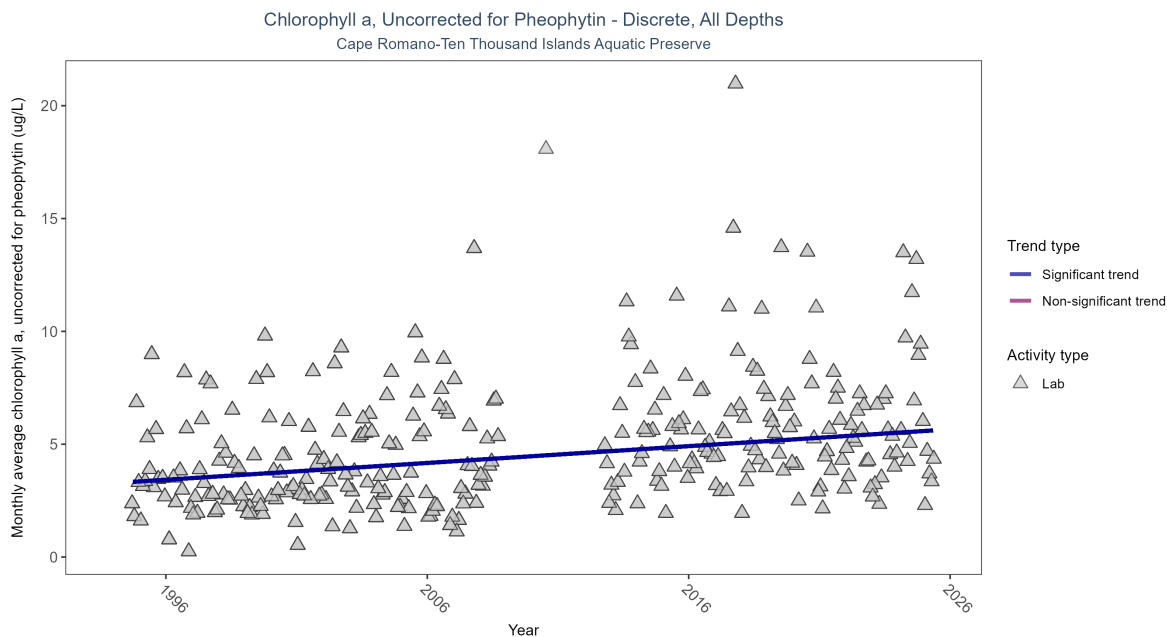


Figure 36: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 19: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	2173	30	1994 - 2025	3.65666	0.27384	3.27391	0.07449	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.07 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

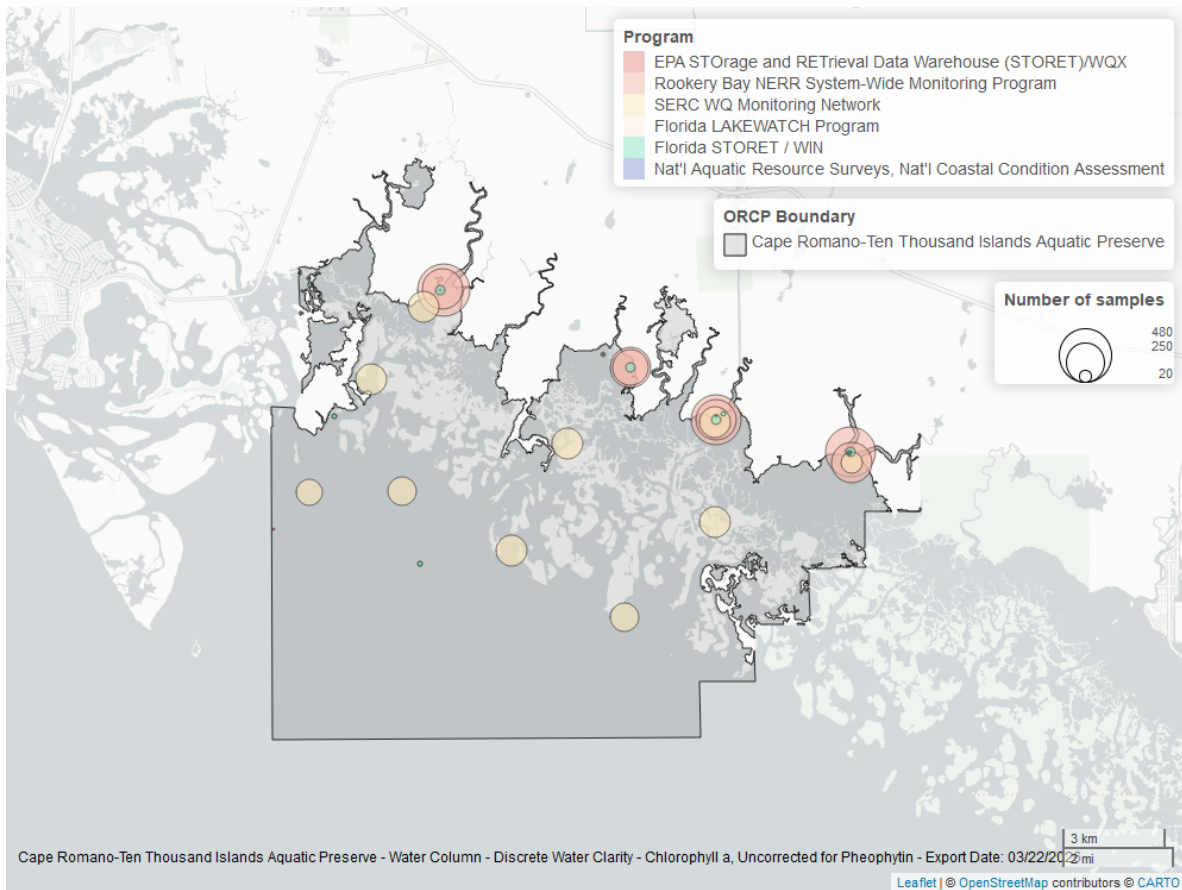


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

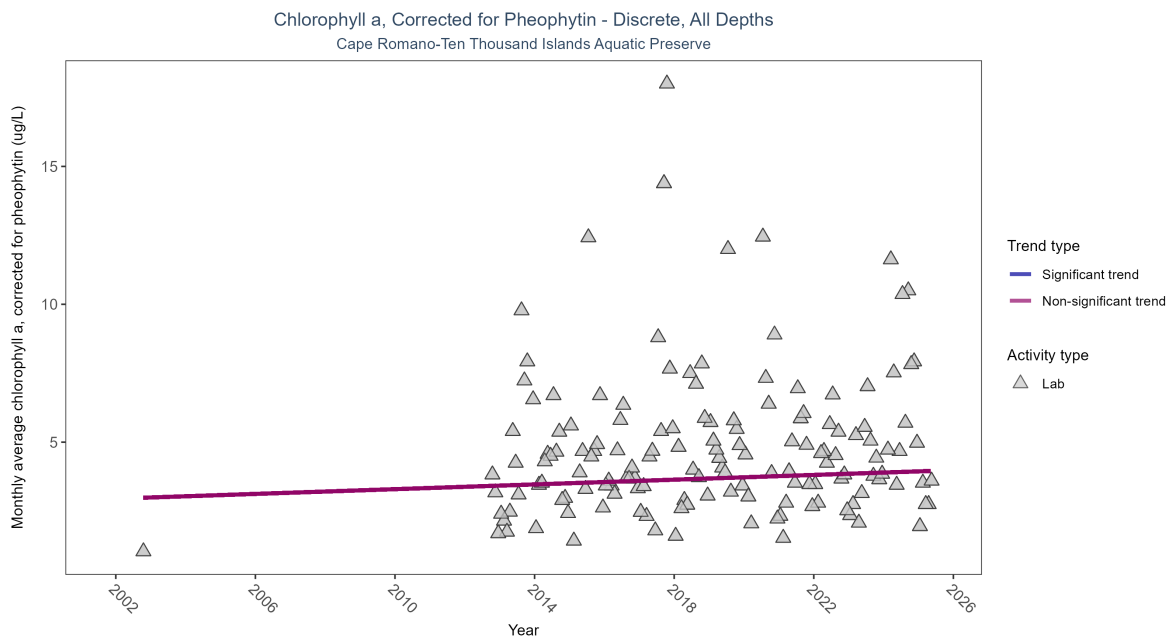


Figure 38: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	639	15	2002 - 2025	3.7	0.09136	2.95078	0.04302	0.1463

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2025.

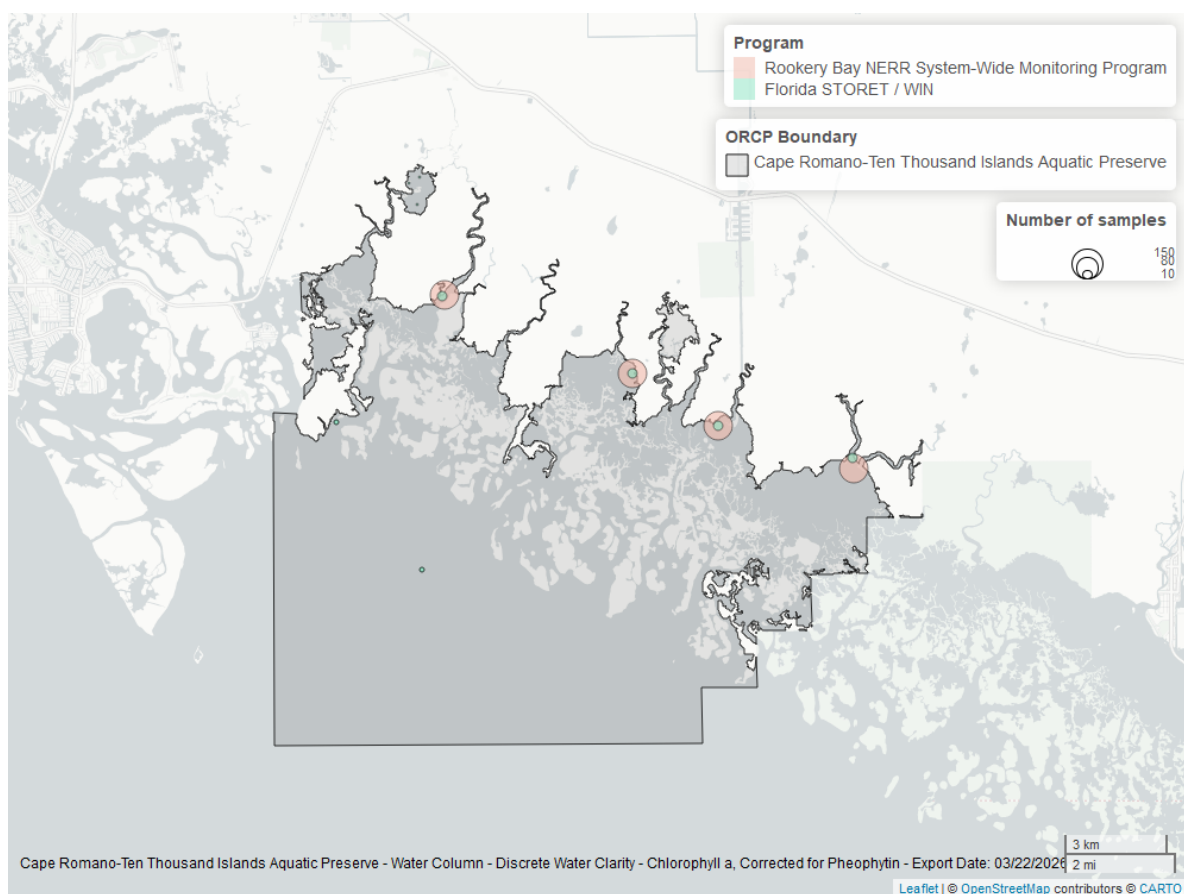


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

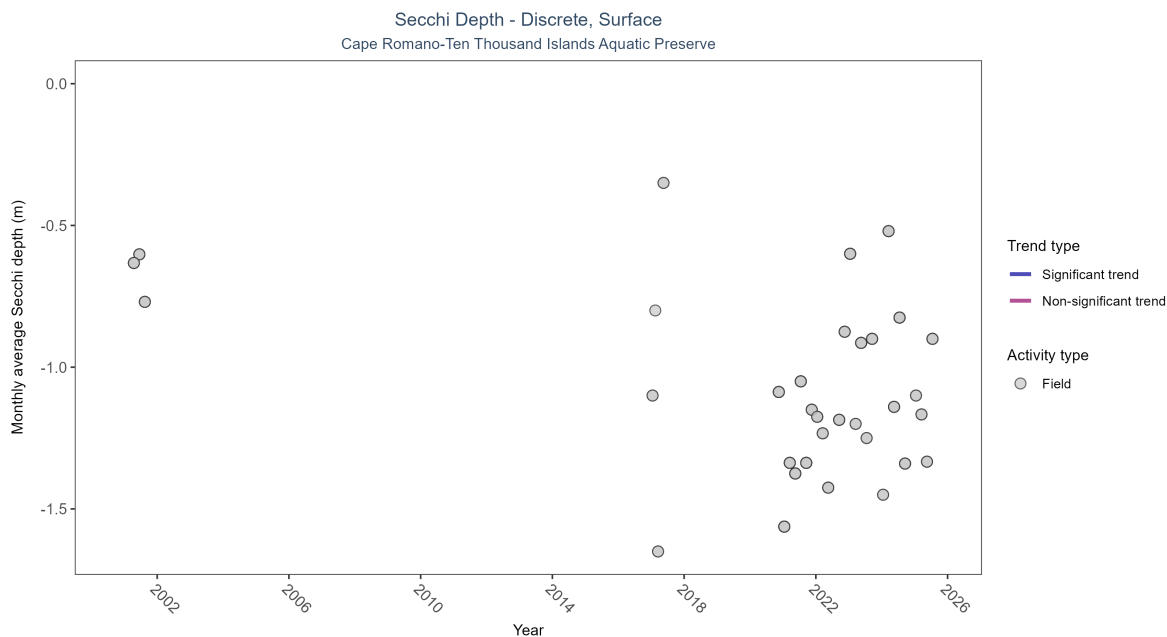


Figure 40: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 21: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data to calculate trend	220	9	2001 - 2025	-0.9	-	-	-	-

There was insufficient data to fit a model for Secchi depth.

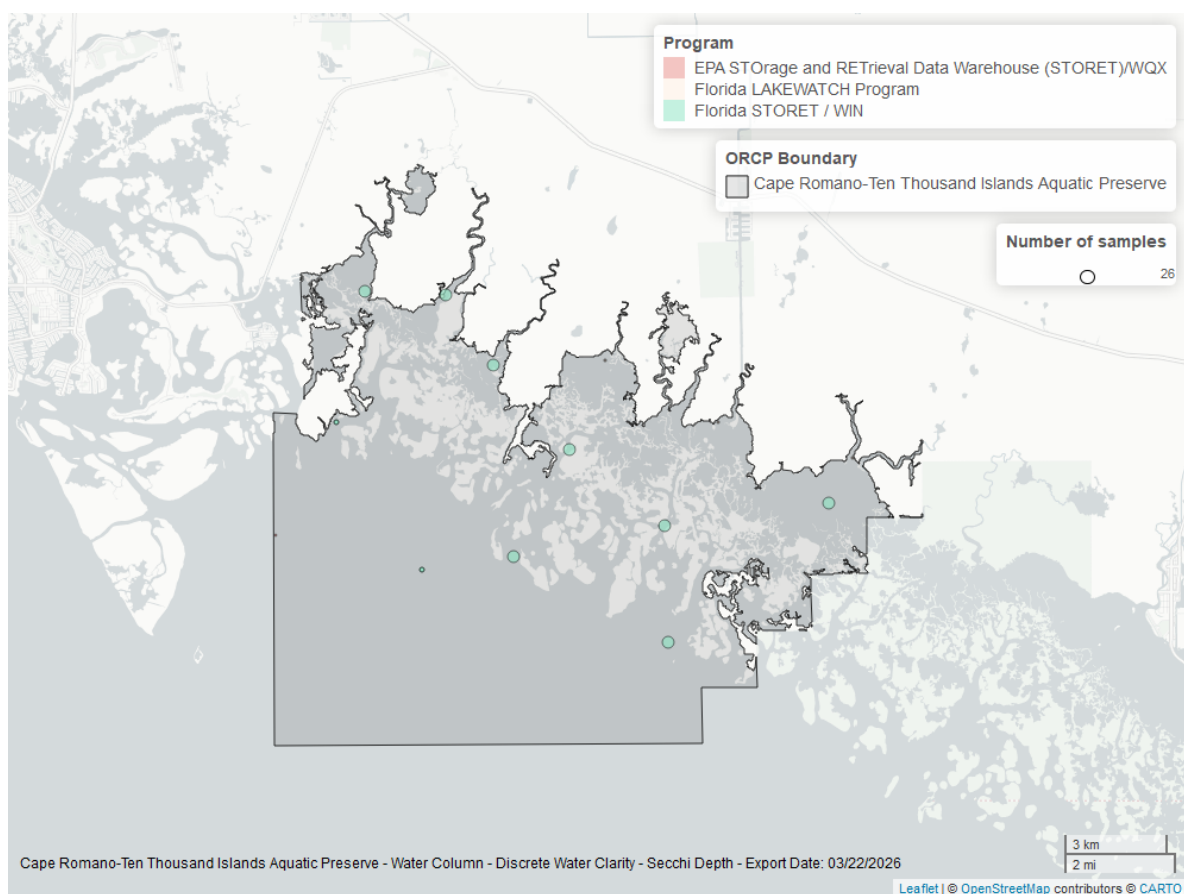


Figure 41: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.